Finding a geodesic current

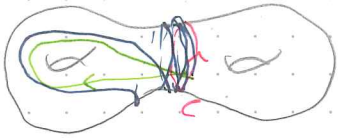
Idea Instead of looking at dense geodesics, work with closed geodesics. Need to deal with weights on the boundary of the intersting rectangles. (possible)

So pick an oriented simple closed curve c_0 on S

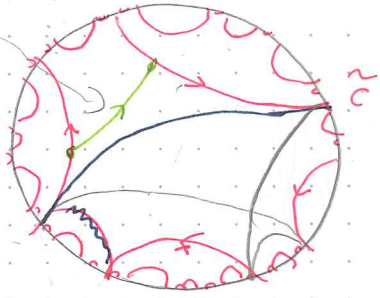
Look at all lifts to $\mathbb{H}^2$
(inital) Endpoints are dense.

Q separating or no

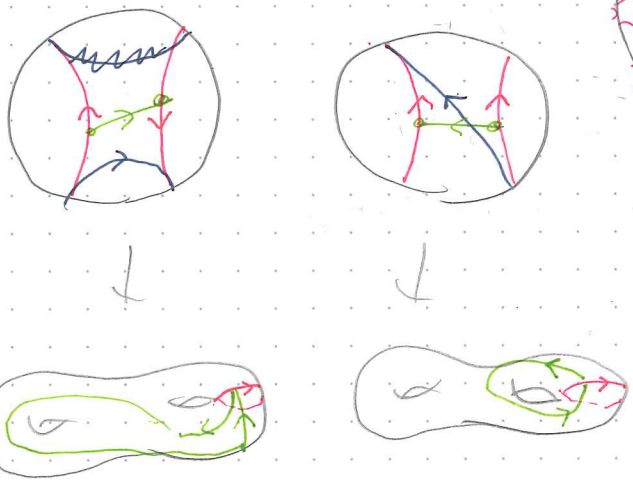
Suffices to find measure of flow boxes w/ endpoints at ends of lifts of c_0.
(Q open or closed?)



Have group elts relating these geodesics



Two cases:

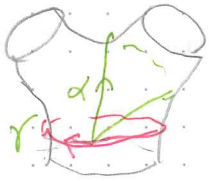


Depend on which side you approach c from

- Do both cases give a well-defined limit in toy model?
- can I restrict to endpoints of one of these types (and still get density)? Probably

compare 2018-09-06 #2

Toy example



$v([\alpha\gamma^n]) \geq n v(\gamma) + v(\alpha)$

(depends on some Dehn twist information)



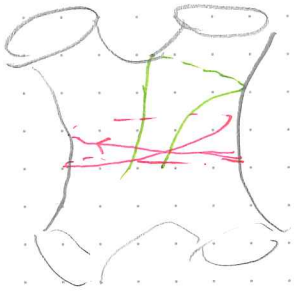
want to show

$\lim_{n\to\infty} v(\alpha\gamma^n) - v(\gamma^n)$ exists

Have ✓ lower bound on $v(\alpha\gamma^n)$

✓ upper bound?

convexity as fcn of n?

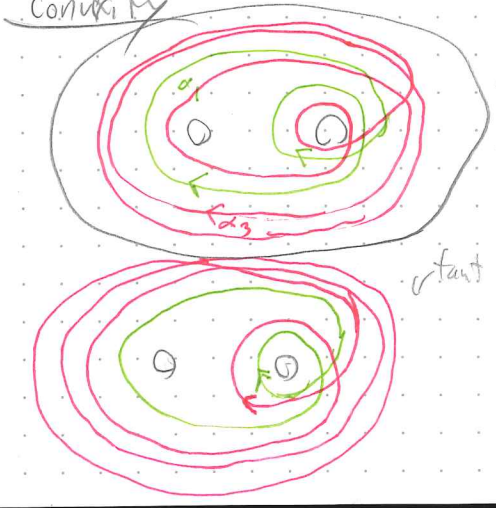




$v(\gamma^n) + v(x) \geq v(\alpha\gamma^{n-1}) + v(\beta)$

Also $v(\gamma^n) + v(x) \geq v(\alpha\gamma^{n-k}) + v(\beta\gamma^{k-1})$ FWIW

Convexity



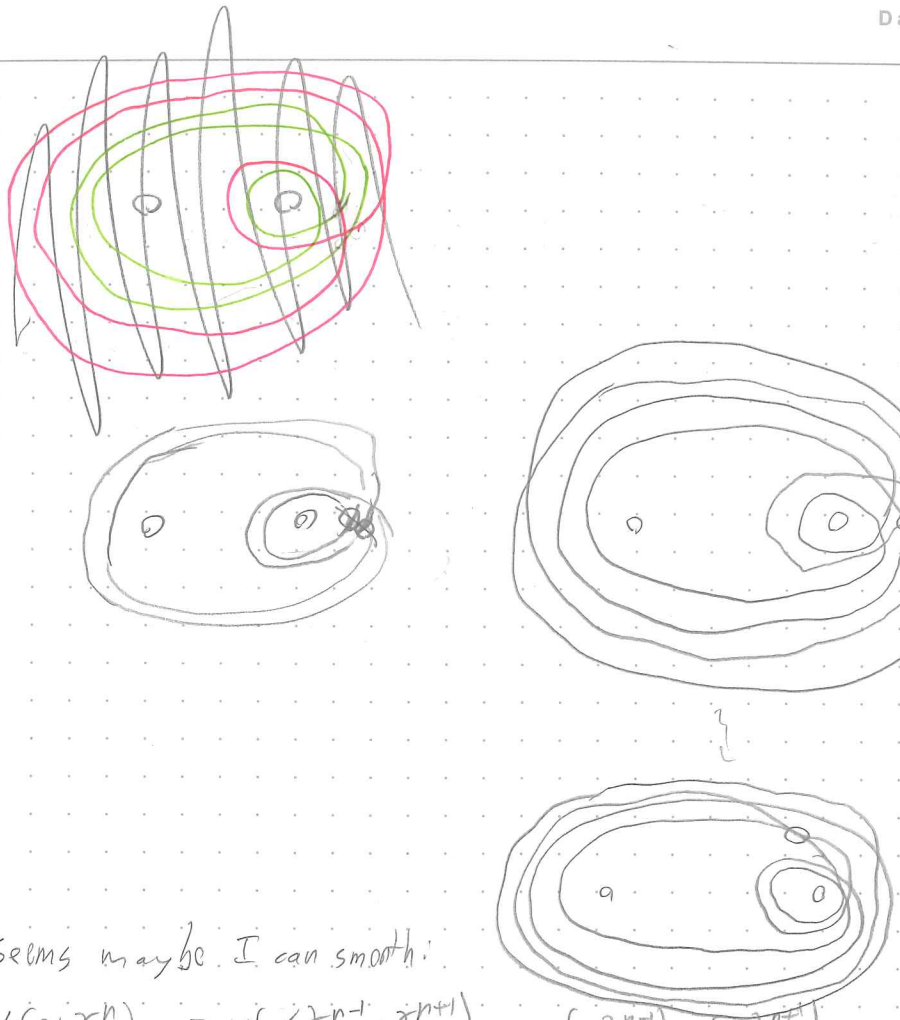
↙ Doesn't seem obvious that convexity is true

↖ not taut

↙ taut

smooths to $v(\alpha^2\gamma^4)$

Not $2v(\alpha\gamma^2)$



Seems maybe I can smooth.

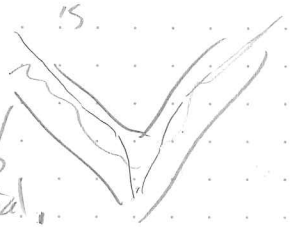
$2v(\alpha\gamma^n) \rightsquigarrow v(\alpha\gamma^{n-1}, \alpha\gamma^{n+1}) \rightsquigarrow v(\alpha\gamma^{n-1}), v(\alpha\gamma^{n+1})$

$\Rightarrow$ concave for n

Need to check this. How does it generalise?

Also, cannot globally be $*$ concave for n!

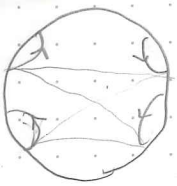
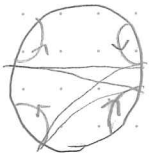
behaviour is



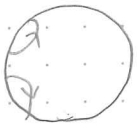
So cases w/ n near 0 must be special.

Note this kind of argument is more delicate - quasi-additivity is not enough.

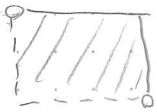
Overall picture several possibilities in a quadrilateral:



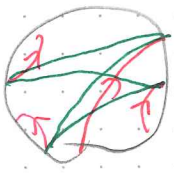
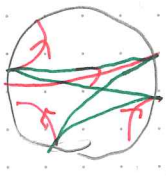





$[a,b] \times [c,d]$?


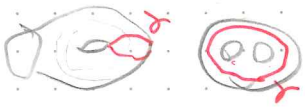
What can there actually be non-zero weights on the boundary of a long chamber?

- There are more cases than about 2^4 cases. Eg

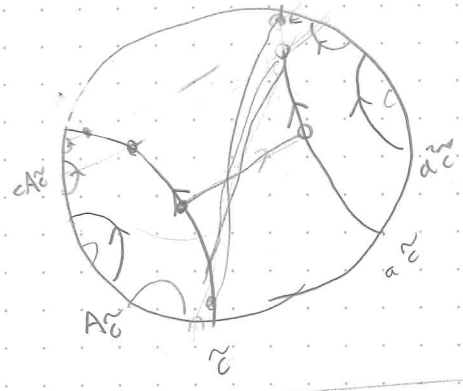
Is this one problematic?

- What cases for α, γ do we need to consider? (intersecting and essential intersections)
  - Pass to cover $\tilde{S}$ corresponding to $\langle \alpha, \gamma \rangle$?
  - Rank 2 subgroup $\Rightarrow$ surface $\tilde{S}$ is pair of pants or punctured torus
  - γ is simple $\Rightarrow$ put in standard form.
  - What can α be?



Is it just $\rho^2 \gamma \beta^{-1}$ — for standard dual generators? Structure of reducible automorphisms of F_2 —

What, exactly, do we measure w/ $\lim_{n\to\infty} v(\alpha\gamma^n) - v(\gamma^n)$?



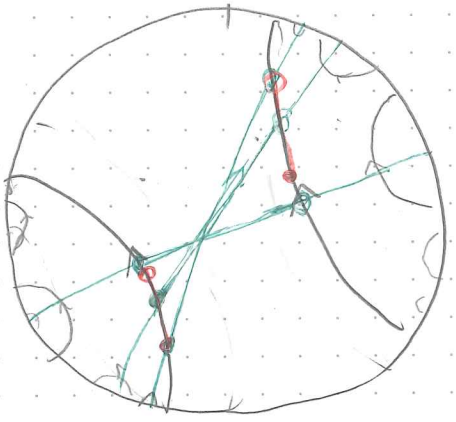
Outer space for F_2: Farey tiling + flaps

Kup, this claim works



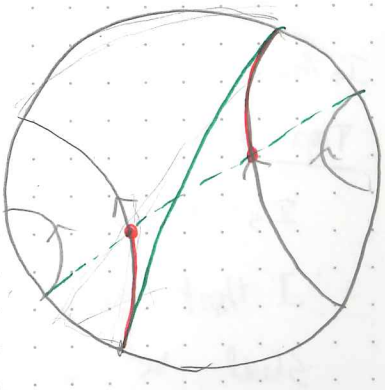
reducible $\Leftrightarrow$ fixes this point

Hmm: seems $Out(F_2) = GL_2(\mathbb{Z})$ acting on $\mathbb{H}^2$?



approximation

Limit!



~~intersection~~ γ (intersections w/ green − intersections w/ red)

Is this finite? Yes

Didac points out:

Additivity. ISSAP

Have weighted curves C_k defined by (smeared) first return map

$a_k = v(C_k)$ $\quad a_{k,\ell} = v(C_k \cup C_\ell)$

convexity: $a_{k,\ell} \leq a_k + a_\ell$

? $a_{k,\ell} \stackrel{?}{\sim} a_k + a_\ell$

stability: ~~$a_{2k} \leq 2a_k$~~ (right?) (wait, no) $\quad a_{k,k} = 2a_k$

smoothing: $a_{k+\ell} \leq v(C_k \cup C_\ell \cup X) \leq a_{k,\ell} + K \leq a_k + a_\ell + K$

isn't this already enough?

$a_{k,\ell} \leq v(C_{k+\ell} \cup X) \leq a_{k+\ell} + K$ ← seems to yield this for continuity

---

More convexity

Suppose you have a weighted directed embedded graph on a surface S

maybe assume no null-homotopic cycles, or else work in some completion

$P(\gamma) = \sum_{c \in \gamma} (w(c))$

Is this log-convex?

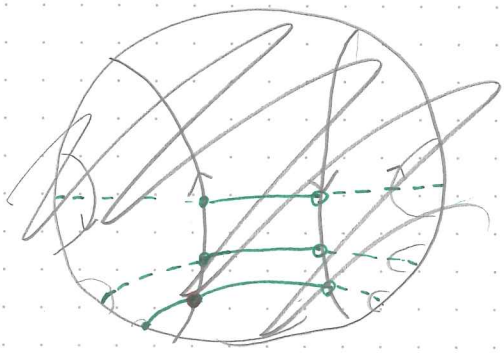
Tropicalise ⇒ length w.r.t. a metric

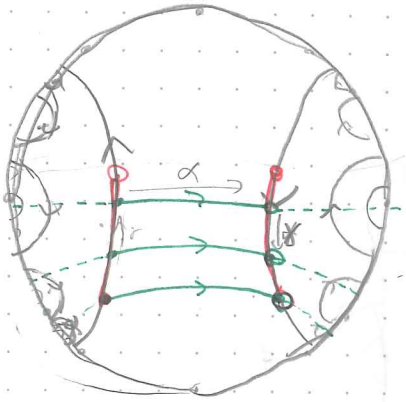
Is ∄ true w/o tropicalization?

I think so.

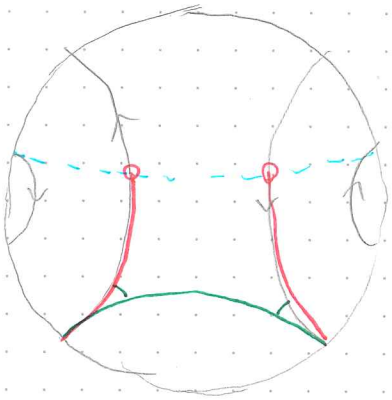
Should be some nice examples.

What is $\lim_{n\to\infty} v(\alpha\beta^n) - v(\beta^n)$ in pair-of-pants case?



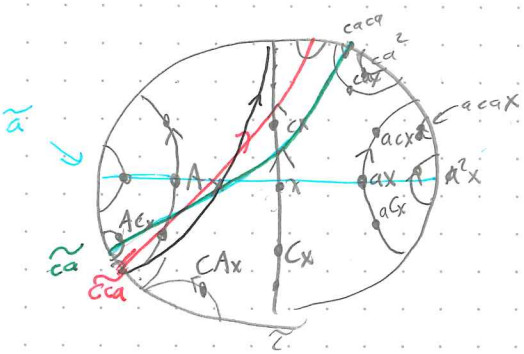


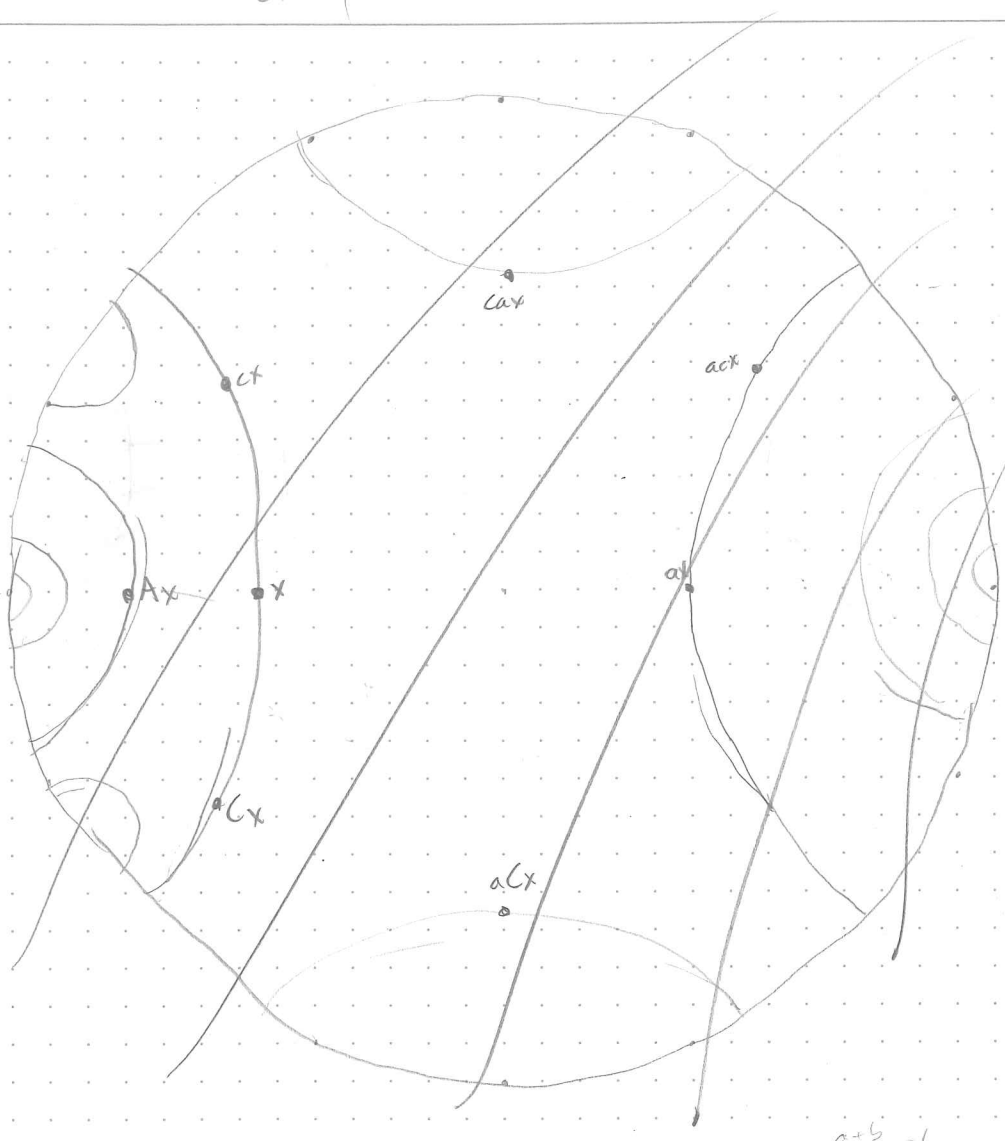
Limit



Something really funny.
Does not seem to work in combinations
Shouldn't these extra bits cancel out?

Ok, lets be more careful.





in $SL(2,\mathbb{R})$: $a(x) = 3x$

$$a(x) = \frac{3x+2}{2x+3}$$

$\frac{a+b}{c+d} = 1 \qquad \frac{a-b}{c-d} = -1$

$a+b = c+d$

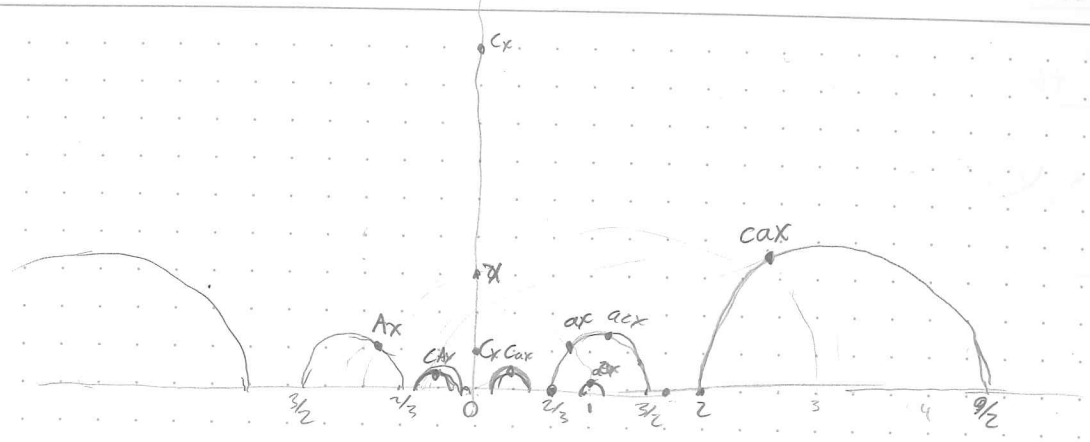
$a-b = d-c$

$a = d \qquad b = c$

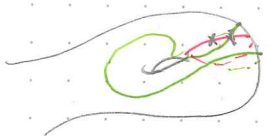
$$0, 1 \mapsto \frac{2}{3} \mapsto \frac{2+2}{4/3+3} = \frac{4}{13/3} = \frac{12}{13}$$

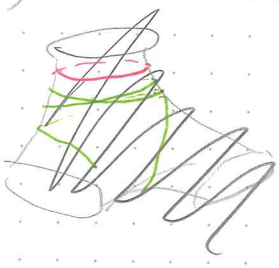
NB $x \mapsto kx$ is conjugate to $x \mapsto \frac{(k-1)x+k+1}{(k+1)x+(k-1)}$

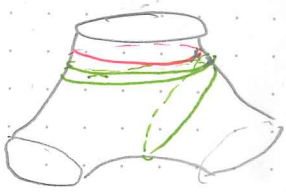
so $x \mapsto \frac{3x+2}{2x+3}$ is conjugate to $x \mapsto 5x$



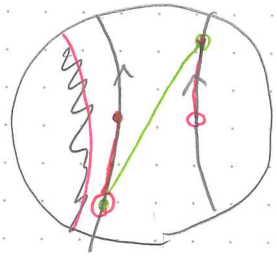
Simpler picture: cut on ξ, lift to $\zeta = 4^z$



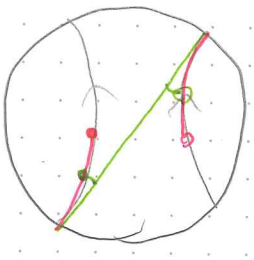




Torus



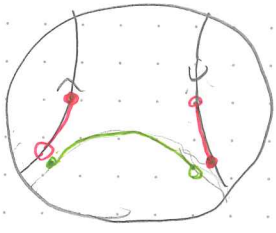
limit

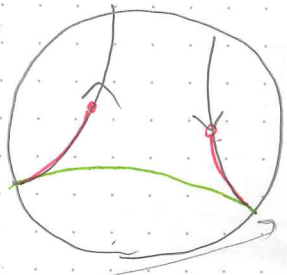


This is strange

Does seem to get very coherent answers
Am I doing the limits right?

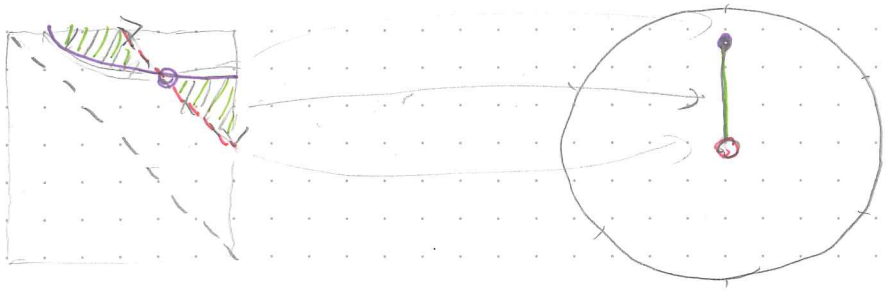
Pants

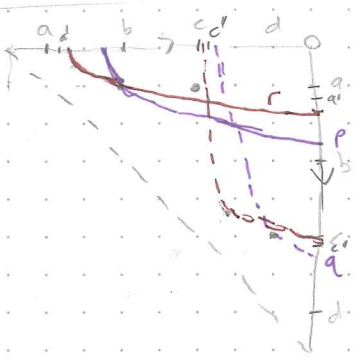


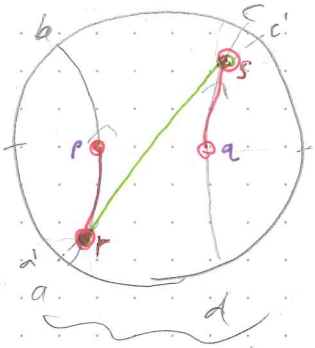


Don't include pencil

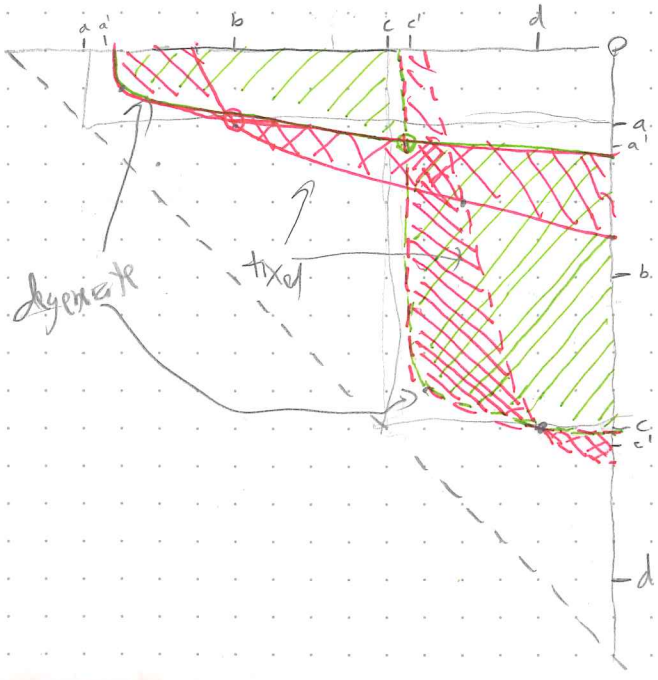
To get the limit right think about subsets of Möbius strip



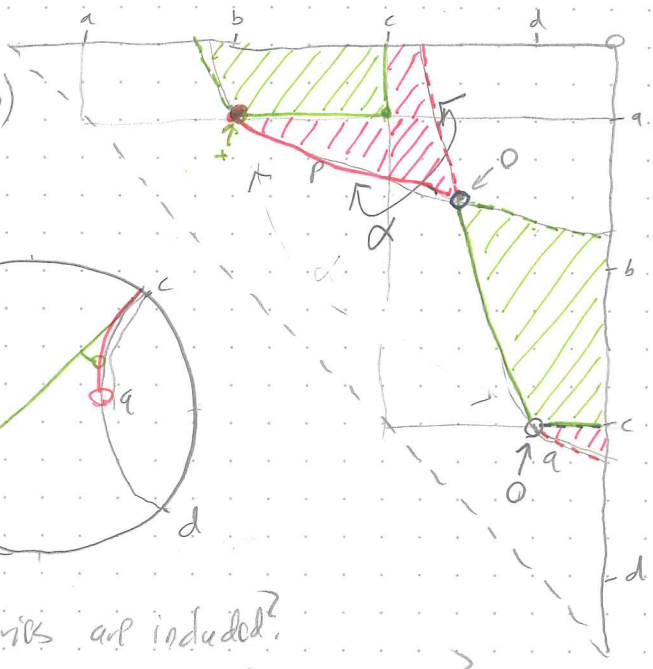




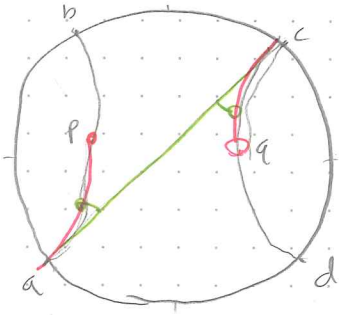
better switch around d to make covered region be positive



Limit (after cancellation)



Torus case

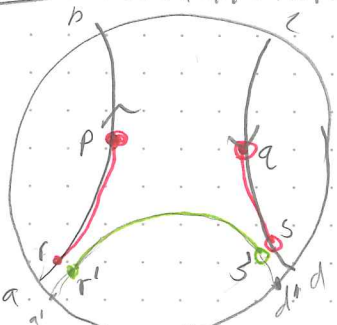


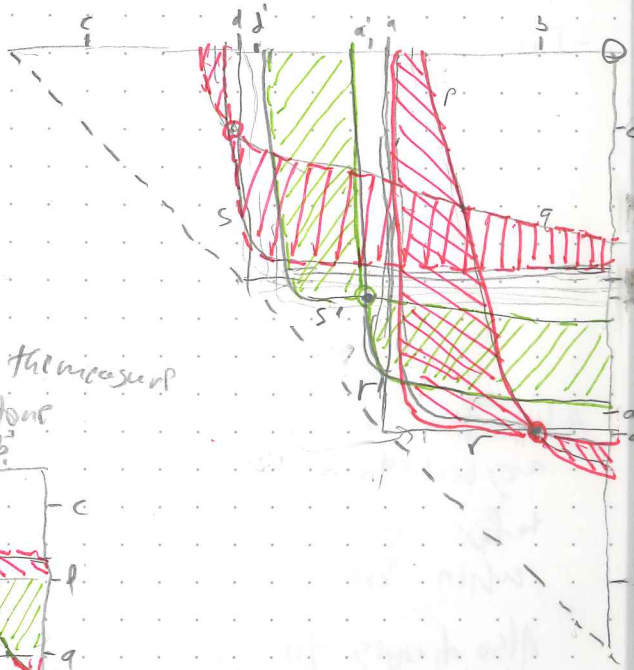
What boundaries are induced?

And do measures actually converge?

Yes converging from above (decreasing fn of n) ← positive region is decreasing, negative region is incr

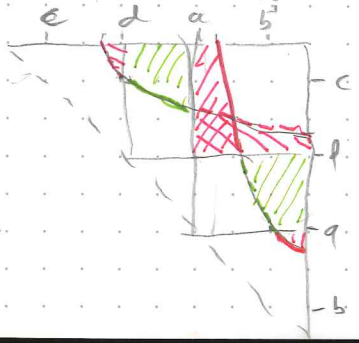
Pants case Seems like there will be a problem from lack of intersection





Issue: sets you are taking the measure of a–c not monotone

Limit



Page: 1

Date: 2018·09·22

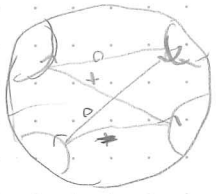
Ok, seems things are better. by Hopf argument (Alexandrov-Fenchel)

measure of the boundary is 0, except maybe for δ-masses on closed curves.

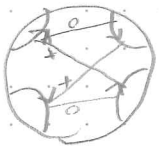
So just need to decide if these bdy pts are included or not.



questionable pts are way out of range. Not included in any counts anyway.

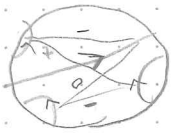


→ not counted



→ do count it!

Probably should but

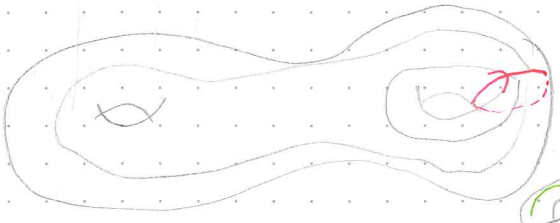


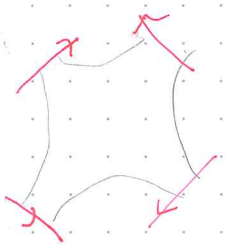
Ah, limit is negative?

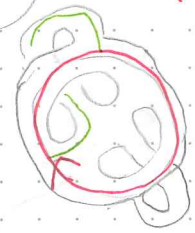
~~Counting is independent~~



- Forgot to check positivity at all. That's clear, smooth that central crossing...
- Finite additivity is really messy for naive rectangles. What does that even mean?





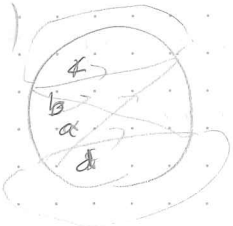


1 bdy
$\chi = -3$
genus 2

Ah hah — for smoothing positivity argument to work, need to fix the pairing. This changes when "crossing" happens.

Also changes the limit argument & makes it harder.

Pick (say)



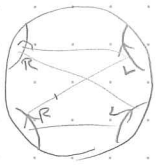
$a \mathbb{C} bD = id$

compute

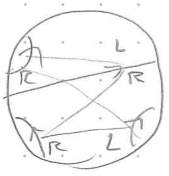
$f(ax^n bx^n) - f(cx^n) - f(dx^n)$

when does δ_x contribute?

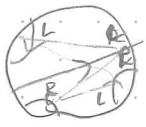
Eg



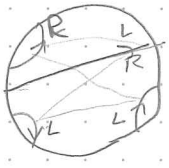
\+ : 2
− : 2
} net contributions clearly 0



\+ : 1 + 1
− : 2



diags: 2 + 1
sides: 1
} contributes



diags: 0 + 1
top/bot: 1

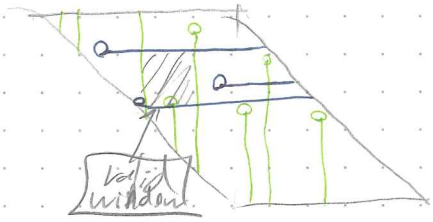
Getting messy. More annoying additivity brackets with this difference between

$v(a\cancel{a}^n b\cancel{x}^n)$ vs $v(ax^n) + v(bx^n)$

Idea
- Restrict which boxes are allowed to avoid "hard" cases
- Prove equivalence of multiple ways of finding the limit



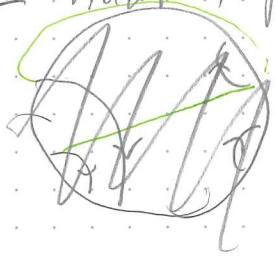
Each geodesic gives an "allowed region" for other endpts

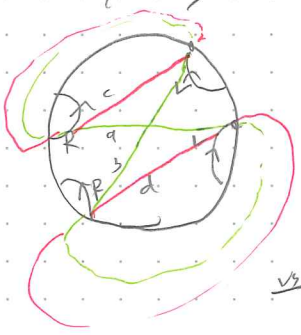


Such rectangles still cover the whole strip. I think.

seems likely that appropriate finite decomposition is true.

So have to prove equality of limits





vs the other one

$$\lim_{n\to\infty} f(ax^n bx^n) - f(ax^n) - f(dx^n)$$

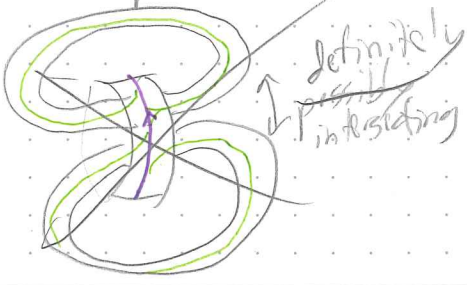
vs

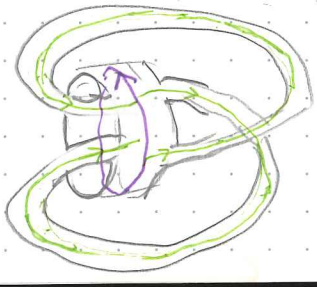
$$\lim_{n\to\infty} f(ax^n) + f(bx^n) - f(cx^n dx^n)$$

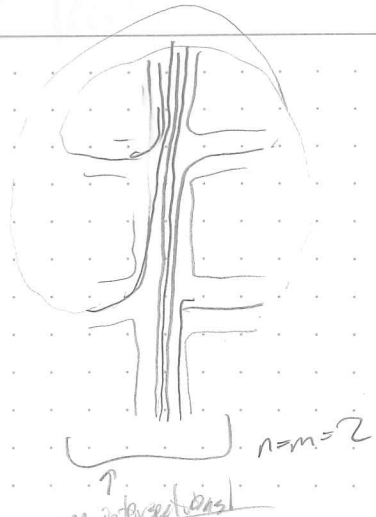
A posteriori, these limits are equal

Can I use some sort of convexity to prove that?

NB All set up in easier-to-deal-with setting, torus-like

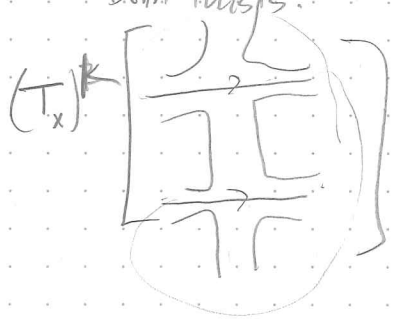






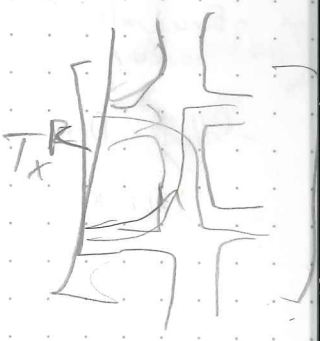
Oh—I should think about Dehn twists.

$(T_x)^k$

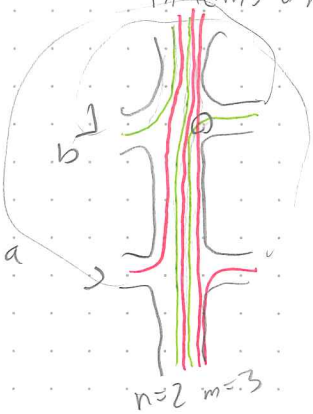


vs

T_x^k



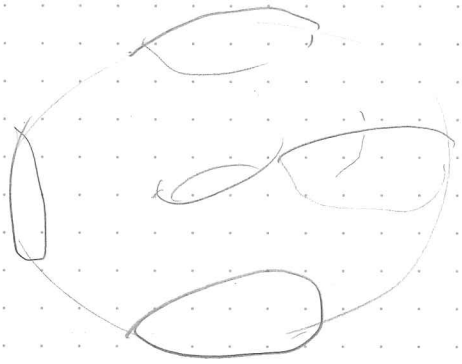
Should also be thinking in terms of joint limits: $\lim_{n,m\to\infty} f(ax^m bx^n) - f(cx^n) - f(dx^m)$



smooths to

$[ax^3 bx^2] \rightsquigarrow [bx^3] \cup [ax^2]$

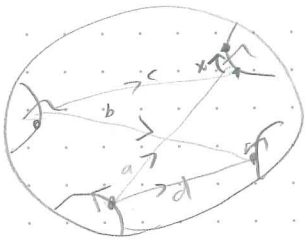
really depends on the details of how twisted a, b are



2018-10-01

What kind of convexity would help?

In setting

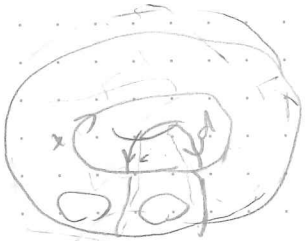


(probably dominated by worry about c/d)
(or maybe better: focus on c/d
minimal model on 2x punctured torus

have $f(n) = v(cx^n)$

$g(n) = v(dx^n)$

$h(n,m) = v(cx^n dx^m)$



know f, g, h change by at most $v(b)$ when a, g, s change

$|f(n) - f(m)| \le |n-m|\, v(x)$

$f(n+1) + f(n-1) \ge 2f(n)$ (convexity) $\Rightarrow$ limit exists

$f(n) + f(m) \ge |n-m|\, v(x)$

[NB: switching orientations]

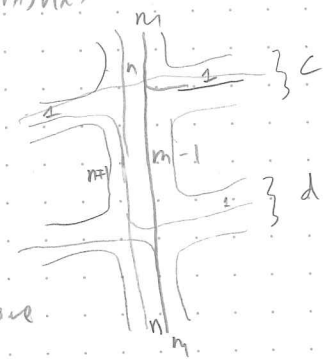
Ok, what about h?

$|h(n_1, m_1) - h(n_2, m_2)| \le (|n_1 - n_2| + |m_1 - m_2|)\, v(x)$

(n, m large, positive)

$h(n,m) + f(0) \ge g(n+m) \sim K_1 + (n+m)v(x)$

$h(n,m) + g(0) \ge f(n+m) \sim$



Know $f(n) \sim K_c + nv(x)$ ← approach from above

$g(n) \sim K_d + nv(x)$

Want $h(n,m) \sim K_c + K_d + (n+m)v(x)$

~~wait~~ ~~$h(n+k, m+k) + f(k) \ge g(n+m+k)$~~ ~~→ oops~~

$h(n,m) + f(k) \ge g(n+m-k)$ in a certain range ($n, m > k$?)

$\Rightarrow h(n,m) \ge K_c + K_d + (n+m)v(x) - \varepsilon$ ← this is nonsense

OK, but really want the other inequality too

Get that by sandwiching $[cx^n] \cup [dx^m]$ (as for convexity)

Need $n \ne m$ to actually have an intersection

Duh, duh $[c] \cup [dx] \searrow [cdx]$

$[cx^n] \cup [dx^{n+1}] \searrow [cx^n dx^{n+1}]$ oh, OK

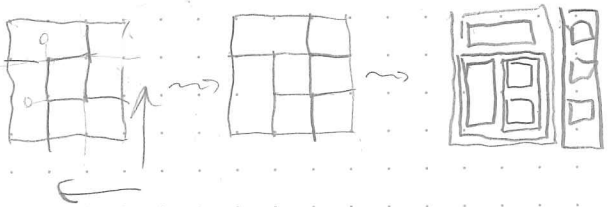
Page: 1

Date: 2018·10·02

Question 1 Is side-by-side additivity enough to prove finite additivity in whisker context?

Question 2 Can you pass to a cover where $c.d.$ both intersect $\mathbb{X}$ just once (so you can do twisting arguments?

Q1 Example:



Given $x, c, d \in \pi_1(S)$ x simple

Consider T = cover of S corresponding to $\langle x, c, d \rangle$

T is a rank 3 surface, if non degen

$\Rightarrow$  or 

In context know x is simple, non separating $\Rightarrow$ T is $T^2 \setminus$ (2 disks)

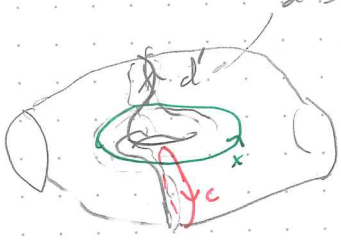
Make x longitude

Q Can we make c, d simple (after pre/post multiplying by powers of x)?

Recall/fixed a question about bases for F_3 with one generator

Is there an analogue for outer space here?

I don't think there's a "basis" thm along the same lines. Eg



not simple. Generates π_1, so have to work in cover.

---

oriented vs unoriented

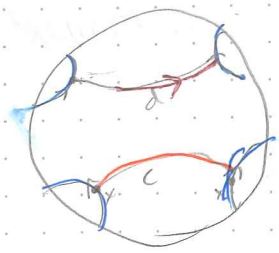
If $\vec{v}$ is an oriented convex fcn, what is $v(\vec{c}+\overleftarrow{c})$?



$v(D(c)) \searrow D(\text{smoothing of } c)$

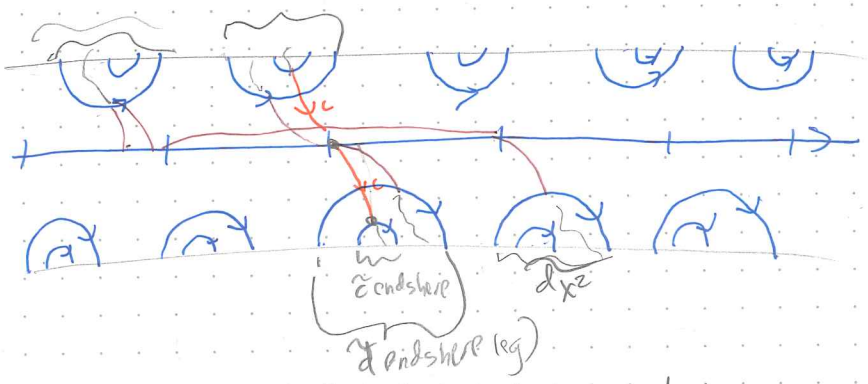
so $v(D(c))$ is convex

Remains to analyze other half $v(\vec{c}-\overleftarrow{c})$. Doesn't seem to be convex.

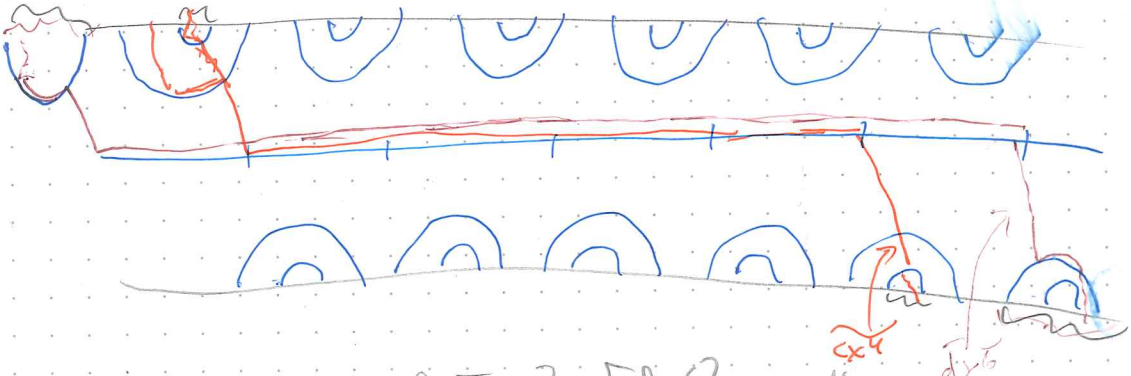


interested in intersections between cx^n & dx^m in nbhd of x.

center window on x
(band model)



dx^2 & c definitely cross: so do cx^n and dx^{n+2}



this union of $[cx^4] \cup [dx^6]$ smooths to

$[dx^5] \cup [cx^5]$

In general with this setup, can smooth

$[cx^n] \cup [dx^{n+2}] \rightarrow [cx^{n+1} dx^{n+1}]$

$[cx^n] \cup [dx^{n+3}] \rightarrow [cx^{n+1} dx^{n+2}]$
or
$[cx^{n+2} dx^{n+1}]$

$[cx^n] \cup [dx^{n+k}] \rightarrow [cx^{n+l} dx^{n+k-l}]$ for $1 \le l \le k-1$

(these cases will vary)

Cool. Looks like this will work for the other smoothing too.

Date: 2018 · 10 · 04

Q. Can I smooth $[cx^n dx^m] \cup [cx^k]$ to $z[cx^l] \cup [dx^r]$

for some exponents? (i.e., preserve orientations in lower bound on $h(n,m)$.

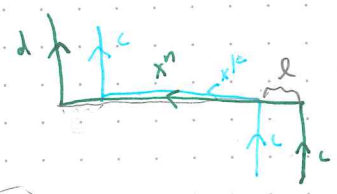
~~knowledge~~ $h(n,m) + f(k) \geq 2f(l) + g(r)$

~~Oops, not good enough. This gives (prob)~~

~~$h(n,m) \leq 1$~~

In fact that's what I need in an oriented case too.

$[cx^n \cdot dx^m] \cup [cx^k]$ focus on c.

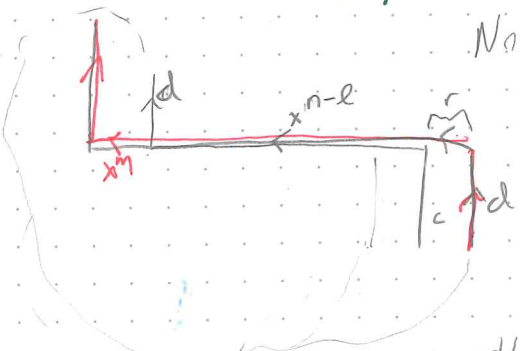


$k<n$ (& const) smooth to $[dx^m c x^{k+l} c x^{n-l}]$

w/ possible values of l depending on $n-k$

Now want to smooth that to

$[dx^{n-l+r}] \cup [cx^{k+l} cx^{m-r}]$



optional with $k+l=m-r$

[or maybe do so

[can also add in some extra pairs of c to get to exact double

Assume $n=m$, can smooth as on previous page

$[cx^n dx^n] \cup [cx^{n-4}] \searrow [dx^n cx^{n+2} cx^{n-2}] \searrow [dx^{n-1}]$ $[cx^{n+2} \cdot x^{n-1}]$

$[cx^n dx^n][cx^{n-4}][x^3] \searrow [dx^{n-1}][[cx^{n+2}]^2]$

Q What's an obstruction for a subdivision rule?
(why haven't I discussed this yet?)

several domains to ask for geometric subdivision rule
- conformal
- Euclidean similarities
- Affine

To prove A convex, additive curve functional comes from intersection #

① Pick a simple closed curve $[x]$, pick basept $p \in [x]$

② For any $c \in \pi_1(\Sigma, *)$ crossing X → define

$$\lim_{n\to\infty} v([cx^n]) - v([x^n])$$

exists.

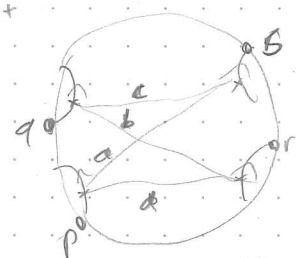
③ For $c, d \in \pi_1(\Sigma, *)$ crossing X (w/ same sign).

$$\lim_{m,n\to\infty} v([cx^n dx^m]) - v([cx^n][dx^m]) = 0$$

④ For $p, q, r, s \in \partial^\infty \tilde{\Sigma}$ suitably arranged

lifts of $\tilde{x}^-$ lifts of $\tilde{x}^+$

a, b, c, d translations relating lifts



set $\mu([p,q] \times [r,s])$

$$:= \lim_{n\to\infty} v([ax^n][bx^n]) - v([cx^n][dx^n])$$

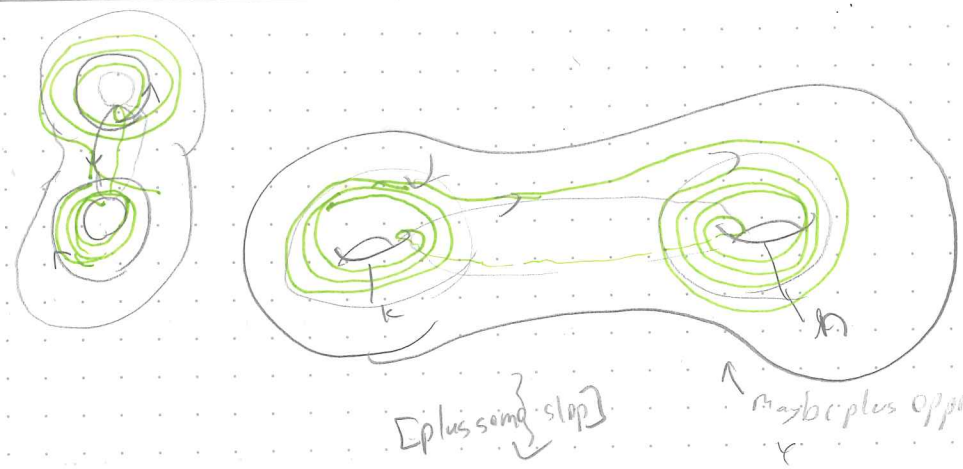
This exists, ≥ 0

⑤ "suitable rectangles" form a ring of sets

⑥

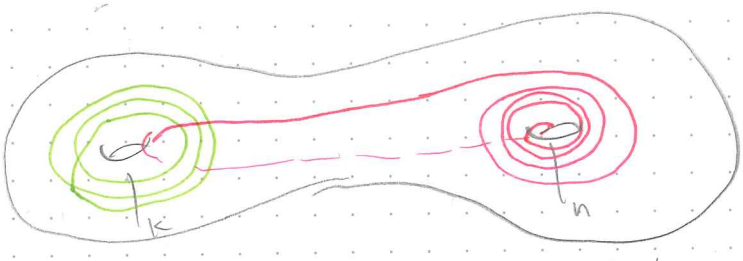
Page: 1

Date: 2018·10·07



[plus some slop]

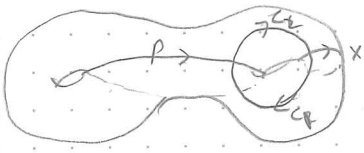
↑ maybe plus opposite spiral



Looks promising actually

To find $\nu(p)$ given the measure at ∞ (w/ winden construction

- Pick curve c crossing both p & X



~~Consider~~ wait

$$\frac{1}{2}\left([c_1 p^k c_2 x^n] + [c_1 p^k c_2 x^n]\right)$$

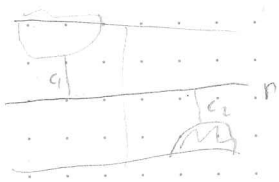
$$-[c x^n] \overset{?}{\sim} k[p]$$

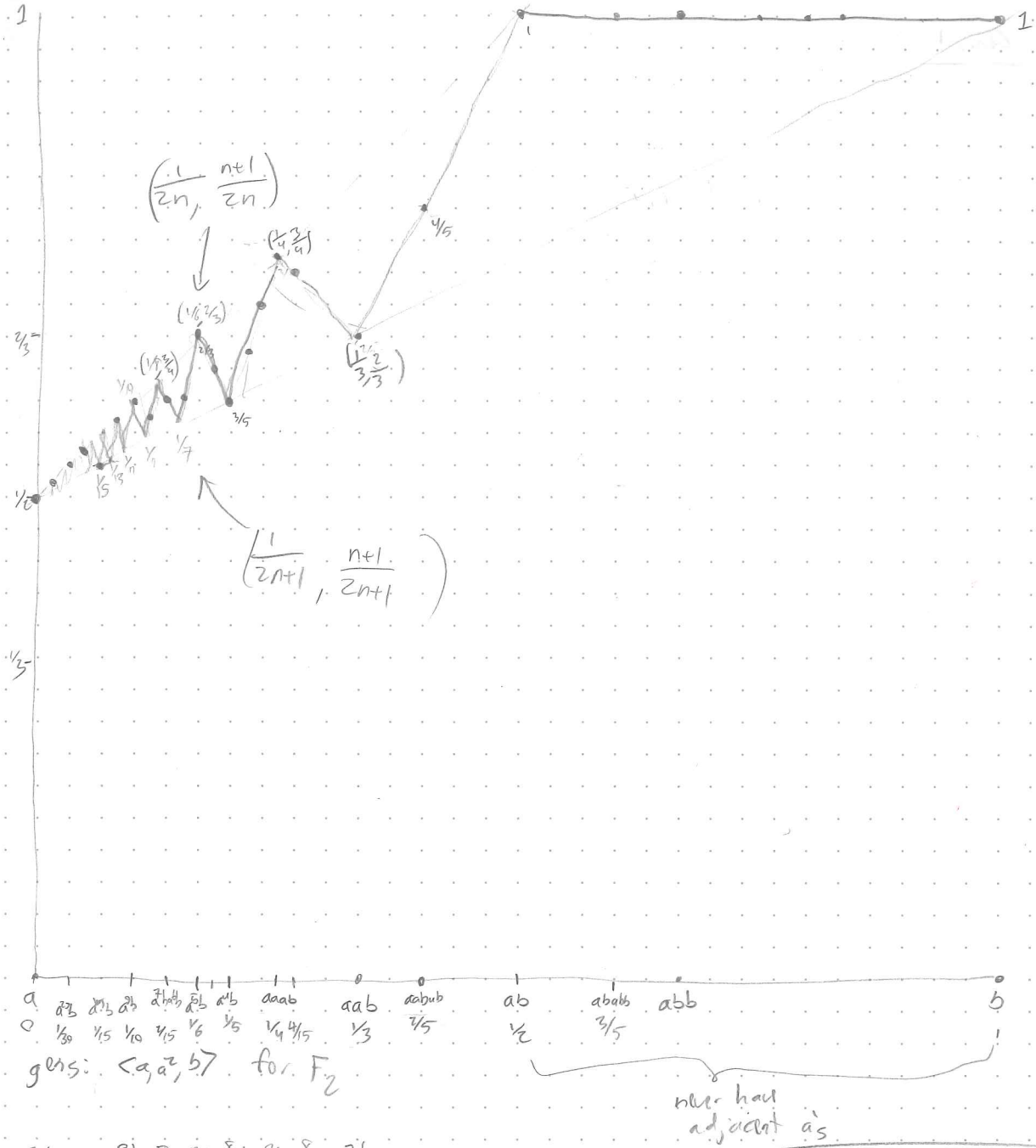


or better:

$$[c_1 p^k c_2 x^n] \sim k[p] + n[x]$$

$$[p^k]\ [x^n][c] \searrow [c x^n][p^k] \searrow [c_1 p^k c_2 x^n]$$





7/60: $a^8b\ a^8b\ a^7b\ a^8b\ a^7b\ a^8b\ a^7b$

3/20: $a^6b\ a^6b\ a^5b$

11/60: $a^5b\ a^4b\ a^5b\ a^4b\ a^5b\ a^4b\ a^5b\ a^4b\ a^5b\ a^4b\ a^4b$

38/60

13/60: $13 \times b$'s $5x\ a^3\ 8x\ a^4$

7/30: $7 \times b$ $5x\ a^3\ 2x\ a^4$

4/15: $4 \times b$ $1x\ a^2\ 3x\ a^3$

Wait. This does not satisfy the bounds

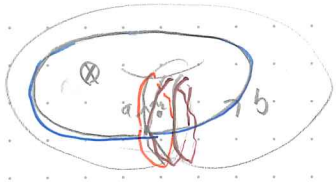
$f\left(\frac{x+y}{2}\right) \leq \frac{f(x)+f(y)}{2} + C|x-y|$

So where did argument go wrong?

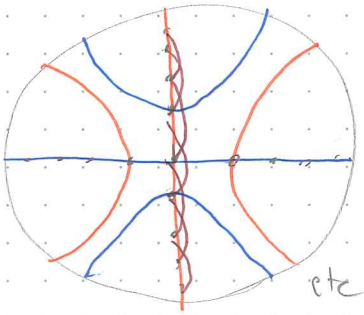
Concretely.

Gens:

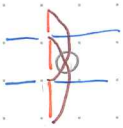
in S:

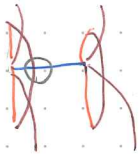


in $\tilde{S}$:



crossings in $\tilde{S}$:

(A)  $K(x) = 4-2 = 2$

(B)  $K(y) = 3-1 = 2$

Claim $\ell(c) \geq \ell(c') - 2$

c' a smoothing of c

Eg $\ell([a^2b][b]) = 3 \geq \ell([abab]) - 2 = 2$

$\ell([a^2]) = 1 \geq \ell([a][a]) - 2 = 0$

$\ell([a^2][a^4b]) = 5 \geq \ell([a^3ba^3b]) - 2 = 6-2 = 4$

Not getting equality?? That's fine, don't really care

~~$\ell([a^{12}b][a^{14}b]) =$~~

$f(\frac{1}{13}) = \overline{x}(\frac{1}{13}[a^{12}b]) = \frac{7}{13}$

$f(\frac{1}{15}) = \overline{x}(\frac{1}{15}[a^{14}b]) = \frac{8}{15}$

$f(\frac{1}{14}) = \overline{x}(\frac{1}{14}[a^{13}b]) = \frac{4}{7} = \frac{8}{14}$

$i(\frac{1}{13}[a^{12}b], \frac{1}{15}[a^{14}b]) = \frac{2}{13\cdot 15}$ → really small

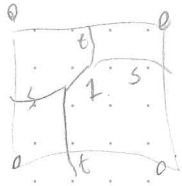
$\frac{1}{14} = (\frac{13}{28})\frac{1}{13} + (\frac{15}{28})\frac{1}{15}$

$f(\frac{1}{14}) = \frac{1}{28}\ell([a^{13}b])$

$= \frac{1}{28}(\ell([a^{12}b]) + \ell([a^{14}b]) + 1)$

$= \frac{13}{28} f(\frac{1}{13}) + \frac{15}{28} f(\frac{1}{15}) + \frac{1}{28}$

↓ way bigger than $i(\frac{1}{13}[a^{12}b], \frac{1}{15}[a^{14}b])$

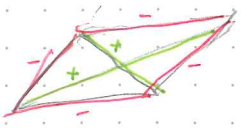


$s+t=1$

Parallelogram law

why does parallelogram law imply additivity?

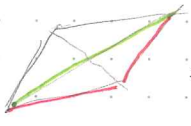
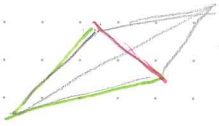
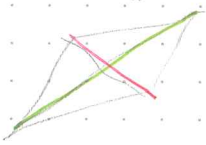
parallelogram law (combs of $\|\cdot\|^2$)

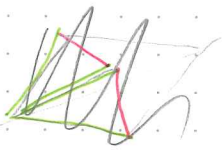


$= 0$

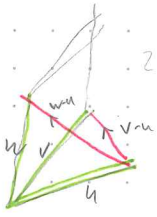
$u+v$
$u-v$ — u
v

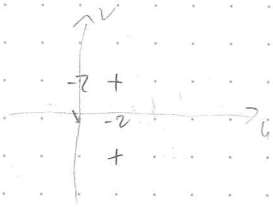
Define inner product

$2\langle u, v\rangle =$  $=$  $= \frac{1}{2}$ 

$2(\langle u, v\rangle + \langle u, w\rangle)$ 

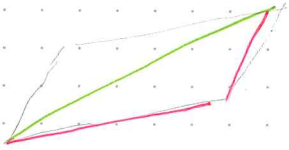




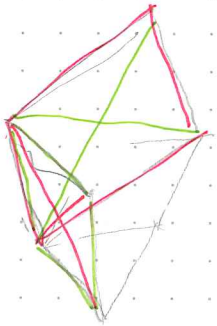


$= \|u+v\|^2 - \|u\|^2 - \|v\|^2 + \|u+w\|^2 - \|u\|^2 - \|w\|^2$

$\|u+v\|^2 - \|u-v\|^2 + \|u+w\|^2 - \|u-w\|^2$

$\langle u, 2v\rangle =$ 

Parallelogram
$(u+w, v)$
minus $(u-w, v)$

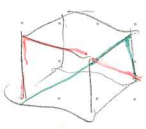


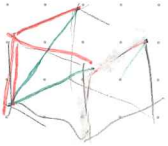
$\|u+w+v\|^2 + \|u+w-v\|^2 = 2\|u+v\|^2 + 2\|w\|^2$
$\|u+w+v\|^2 + \|u-w-v\|^2 = 2\|u-v\|^2 + 2\|w\|^2$

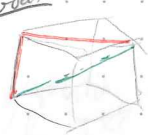
Not really reasonable.

OK, just punt.

Goal







Q (1) What is EL on a non-simple graph?

Is it quasi-convex?

(2) What about CFP version of coarse EL? (modified to be slightly less coarse) ↪ homotopy

(1) $EL(c) = \sup_{\rho} \frac{l_\rho(c)^2}{A_\rho}$



How to compare $l_\rho(c)$ and $l_\rho(c')$?

(since we don't know ρ apriori)

can still do it.

Normalize: $A_\rho = \sum \rho(e_i)^2 = 1 \Rightarrow \sum \rho(e) \le N$ (comparison of norms)

There is some bound: every cross-switch connection in $\tilde{\Gamma}$ has max # of times it covers other edges K

Then $l_\rho(c') \le l_\rho(c) + NK$

wait now we need to square this. Not precise enough

No this is good-interested in $\sqrt{EL}$

so this also extends

Vertex-based notion of EL: this also extends

~~Since~~ For embedded ~~planar~~ graphs: just smooth, get inequality directly

For immersed graphs: Same arguments & bounds gives additive smoothing

Eg What's the extremal length wrt non-simple generating set? $\langle a, a^2, b \rangle$ for $\mathbb{Z}^2$

I think these inequalities were going the wrong way, no?

$EL = \sup \frac{\ell^2}{A}$ want $\sqrt{EL[c]} \geq \sqrt{EL[c']} - K$

Need to start w/ an optimizer for c', not for c



~~Soon~~ OK, we're good:

Pick optimizer for c': $\sqrt{EL[c']} = \frac{\ell_p[c']}{\sqrt{A_p}} = \ell_p[c']$

Normalize $A_p = 1$

$$\sqrt{EL[c]} \geq \ell_p[c] \geq \ell_p[c'] - NK$$

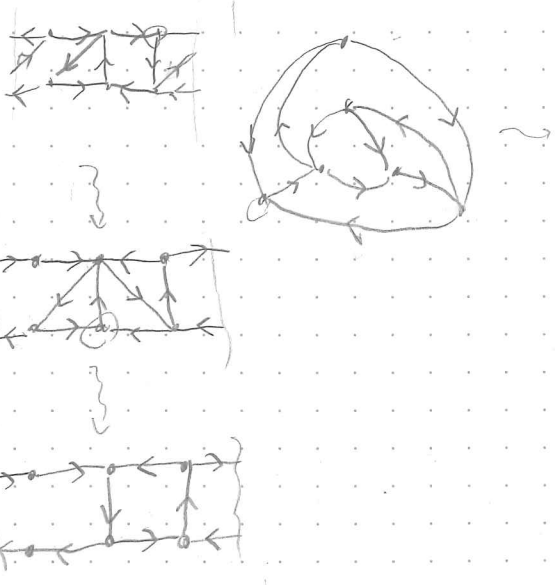
What about $SL(3)$ positivity?

Key case: Annulus

this cluster alg.



(rank 6)



What are the imaginary roots of this cluster algebra?

Wait, back up. In SL_2 case, the algebraic thing is $\operatorname{tr} A = \lambda + \lambda^{-1}$

+ Take log to make additive instead of multiplicative

- Does not satisfy smoothing; things are sign-definite, not always positive

- stable version is $\lim_{n\to\infty} \frac{1}{n} \log(\operatorname{tr} A^n) = \lambda$

- For $SL(3)$, similarly get $\lim(\quad) = \lambda_1$ $\quad \left(\lambda_1, \lambda_2, \lambda_3\right)$ $\pi = 1$

Ok! So that is the right characteristic fcn

Now Does this satisfy the smoothing inequality?

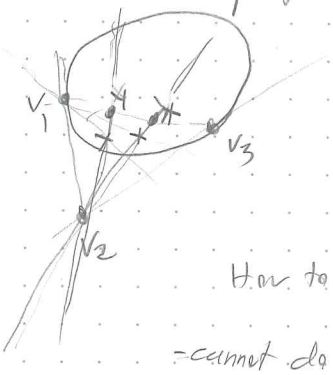
What's the class of reps. we are looking at?

"anosov" → eg: int map $\partial\widetilde{\Sigma} \to \mathbb{RP}^n$ (or better to flags in $\mathbb{RP}^n$)
or connected component of image of Fuchsian reps

$\widetilde{\pi}_1(\Sigma) \to SL(2) \to SL(n)$

To prove Maybe pass to cover generated by crossing?

For $SL(3)$ Convex projective structures $\Rightarrow$ get an asymmetric metric?



Matrix: $A(p v_1 + q v_2 + r v_3) = p\lambda_1 v_1 + q\lambda_2 v_2 + r\lambda_3 v_3$

For a geodesic $x = \{s v_1 + t v_3\}$

$y = \{s\lambda_1 v_1 + t\lambda_3 v_3\}$

How to extract λ_1? (not λ_1/λ_3)

- cannot do it just from behaviour on $v_1 - v_3$ line (projectively)

2018-10-19

Push-forward on currents

Given η a geodesic current on Σ_1

is there a "push-forward" current $f_*\eta$ on Σ_2? ($f: \Sigma_1 \to \Sigma_2$)

(what about pull-back?)

Issue: For many interesting cases, need to allow punctures/boundary

Probably not a big issue?

Idea: Define a fcn on curves on Σ_2 (intimum over all lifts to Σ_1)
check: satisfies additivity, smoothing

$\Leftrightarrow$ get out a current by our theorems

Page: 1

Date: 2018 · 12 · 14

Question: Return maps

How does the averaging work?



$s+t=1$

Stat (Iterate)	0	½	1	1½	2	3	4
0\0	1						
1\0	1						
0\1			1				
1\1			s		t		
0\2				s		t	
1\2				s	st	t^2	
0\3					s	st	t^2
~~1\3~~					s^2	$2st$	st^2 t^3
0\4						s^2	$2st$ st^2 t^3
1\4						s^2	$2st$ $3st^2$ st^3 t^4

cyclic

Val (multiple of f(c))

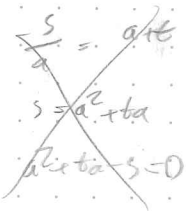
0 — 5

$s\cdot\frac{1}{2}+s\cdot\frac{1}{2}+t=1$

$s^2+s+2st\cdot\frac{3}{5}+2t^2=2$

Ok, good





Stationary distribution is not obvious (so avg length not clear).

oh, it is clear, no?



Good. avg ~~length~~ progress is then t

$\frac{1}{1+s}\left(\frac{1}{2}s+\frac{1}{2}st+t\right)=\frac{1}{1+s}$

as expected

Claim.

Average length grows like $\frac{n}{1+s}$

Length of M^n (with s_i initial weights)

	s_0	$1\cdot 0$
$n=0$		
$n=1$	$\frac{1}{2}s$	$\frac{1}{2}s+t$
$n=2$		

Hmm not enough info to make this obvious

stronger claim?

General situation

unsmeared first return dots

g $x_0 \to x_1 \to x_2 \to \cdots \to x_{n-1}$

Smearing: weight a_i on x_i; at least one a_i is 1

Lemma 1 $\sum a_i x_i$ is a stationary distribution

Proof $f(\Sigma a_i x_i) = \sum a_i a_{i+1} x_{i+1} + \sum a_i (1-a_{i+1}) f(x_{i+1})$ (by def)

$= \cdots =$

$= \sum_k a_k x_k \left(a_{k-1} + (1-a_{k-1}) a_{k-2} + (1-a_{k-1})(1-a_{k-2}) a_{k-3} + \cdots \right)$

$= \sum a_k x_k$. ∎

Let $C_{i,k}$ = closure of $g^k(x_i)$

Lemma 2 $\nu(C_{i,k}) \simeq \frac{k}{n} \nu(C)$ Pf smoothings, homogeneity

---

Let $r(x_i)$ = expected # of spaces to move from x_i

so $r(x_i) = a_{i+1} + (1-a_{i+1})(1 + r(x_{i+1}))$

Lemma $r(\Sigma a_i x_i) = n \Sigma a_i$

Pf $r(\Sigma a_i x_i) = \sum \left[a_i a_{i+1} + a_i (1-a_{i+1}) (1 + r(x_{i+1})) \right]$

$= \sum \left(a_k a_{k-1} + 2 a_k (1-a_{k+1}) a_{k-2} + 3 a_k (1-a_{k-1})(1-a_{k-2}) a_{k-3} + \cdots \right)$

Hmm. Not clear.

Date: 2018.12.17

Return maps

(cont. from 2018-12-14 #3)

Cor of Lemma 1: From the distribution $\sum a_i x_i$, the total weight of paths arriving at any given x_k (stopping or continuing) is equal to 1.

[**Pf** This is just a restatement of the result, since the probability of stopping is equal to a_k.

Prop $r(\sum a_i x_i) = n$ [Expected # spaces to move]

Pf We're looking at ~~weighted~~ # of segments covered.

By Cor, the ending points of covered segments are equally covered with weight 1.

It's now pretty immediate that

$M^r(\sum a_i x_i)$

is a weighted sum of $b_{i,k} c_{i,k}$, with $\sum b_{i,k} \cdot \frac{k}{n} = r$.

This is what we needed.

→ **Better Proof Directly:**

Prove by induction on # steps since an $x_i \in T_0$ (i.e., $a_i = 1$)

Clear if $a_{i-1} = 1$

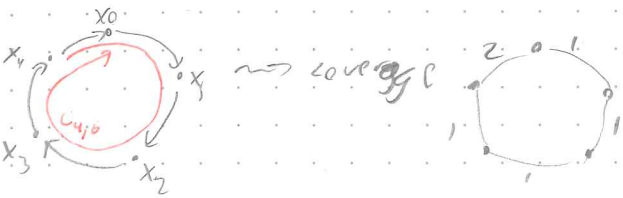
Otherwise, by induction net inflow to x_{i-1} is equal to 1,

so net outflow is $(a_{i-1}) + (1 - a_{i-1}) = 1$

(a_{i-1}: starting at x_i; $1 - a_{i-1}$: continuing from earlier trajectories.)

To get better bounds on smeared first return weights.

Think about the orbits & degree to which each segment is covered, in this sense:



[Need more precise terminology here]

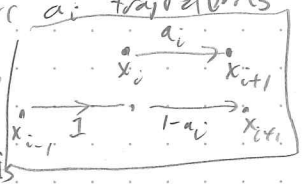
Also set $a_i = \varphi(x_i)$ weights
by assumption that τ_0 is complete, some $a_i = 1$
WLOG, set this to be a_0

Lemma Smeared first return map from the weighted collection $\sum a_i x_i$ gives a weighted collection of curves $\sum w_{ik} C_{i,k}$ so that total coverage of each segment is 1.

Proof Prove this for each segment $x_i - x_{i+1}$ by induction on i.

For $i=0$, this is clear, since $a_0 = 1$, and no earlier trajectories continue through x_0.

For $i>0$, have a_i trajectories starting at x_i and going to x_{i+1}. Also have



weight 1 of trajectories arriving at x_i from x_{i-1}, and so weight $1-a_i$ continuing on to x_{i+1}.

Corally As above, for nth return map, cover each segment with weight n.

Proof Use fact $\sum a_i x_i$ is stationary.

Lemma If we have a weighted collection of curves $\sum w_{i,k} c_{i,k}$ so that total coverage is constant at n, then we can split up the sequence into a finite sum.

convexity of $\sqrt{EL}$

To prove

$$\sqrt{EL[C_1+C_2]} \le \sqrt{EL[C_1]} + \sqrt{EL[C_2]}$$

↑ optimum metric — Normalize

$$EL = \sup_p \frac{l_p^2}{A}$$

$$l_p(C_1+C_2) = l_p(C_1) + l_p(C_2)$$

$$EL(C_1) \ge \frac{l_p(C_1)^2}{A}$$

$$\sqrt{EL(C_1)} + \sqrt{EL(C_2)} \ge \frac{l_p(C_1)+l_p(C_2)}{\sqrt{A}} = \sqrt{EL(C_1+C_2)}$$

Can it be that easy?

I guess so. Huh.

10 (2019-02-14)

What should p-EL be?

Want $p=\infty \rightsquigarrow$ length

$p=2 \rightsquigarrow$ EL

$p=1 \rightsquigarrow$??

For graphs

$$E'_p[C] = \sup_{\rho: E\to\mathbb{R}} \frac{\ell(C;\rho\alpha)}{E^p_\infty(\rho)}$$

Graphs $E_p[C] = \sup_\sigma \frac{\ell(C;\sigma g)}{\|\sigma\|^p}$

Clearly convex, union, homogeneous, stable, smoothing ~~convex union~~

What happens for $p=\infty$? $p=1$?

$p = \infty$

~~There's some shortest path on g~~

Let σ be optimizing scale (map), γ be minimizing curve rep.

Then $\ell_{\sigma g}(\gamma) = \int_\gamma \sigma \, dx \leq \|\sigma\|_\infty \ell_g(\gamma)$

so may take $\sigma = 1$, γ that a g-geodesic

~~Inequality goes the wrong way~~

$p = 1$

~~Is this~~

$$\sup_\sigma \frac{\ell_{\sigma g}[c]}{\int_\Sigma \sigma \, dg}$$

How does this relate to $\inf \{ \ell(D) \mid i(\zeta, D) \geq 1 \}$ (over weighted curves)

~~Given such a D in this set~~ or $\inf_D \frac{\ell(D)}{i(\zeta, D)}$

$\ell_{\sigma g}[\gamma]$

What is this for simple multi-curves?

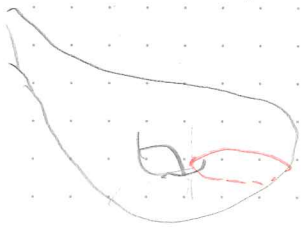
Must be an annular structure.

$p = \infty$ | Better

Let γ = optimal rep in g, σ optimum metric

Then $E_\infty^*(c) = \frac{\ell_{\sigma g}(c)}{\|\sigma\|_\infty} \leq \frac{\ell_{\sigma g}(\gamma)}{\|\sigma\|_\infty} \leq \frac{\|\sigma\|_\infty \ell_g(\gamma)}{\|\sigma\|_\infty}$ as desired

OK, so this is maybe a reasonable notion. But $p=\infty$ case is still really confusing



Hyperbolic geodesic is not the Euclidean geodesic (eg forbidden region around cusp) for simple hyperbolic geodesics)

So what does optimal metric look like for large p?

Must • have shortest curve going through each point
• in conformal class
• hyperbolic geodesic needs to be nearly geodesic

Usual argument: reduce metric in regions not covered → reduces length

Do not need evenly covered

Metric cannot be one from quadratic differential ??

This is fine: Metric is something w/ annuli of varying width.
eg $ds^2 = dx^2 + f(x)\,dy^2$ / $(x,y) \sim (x+1, y)$
cake surface of revolution



Q Is this the only (typo) metric?

Q should I contemplate discontinuous metrics

Back to E_1 metric:

$$E_1(C) = \sup_\sigma \frac{\ell(C;\sigma_g)}{\|\sigma\|_1}$$

Q Is this

$$\sup\{i(C,D) \mid D \in \mathcal{C}, \ell(D) \le 1\}?$$

NB. This is convex.

[this is like min-cut for C, so E_1 is like max-flow]

Given such D: Let d be geodesic rep

$$\sigma_d = \chi(d \times \varepsilon)$$ characteristic fcn on ε-nbhd of d (scaled by weight)

Then

$$\frac{\ell(C,\sigma_d)}{\|\sigma\|_1} \sim \frac{\varepsilon\, i(C,D)}{\varepsilon\, \ell(D)} = \ldots$$

as desired

But how do you know you can't do better?

This is on the same side:

Claim

$$E_1(C) = \sup_\sigma \frac{\ell(C;\sigma_g)}{\|\sigma\|_1} = \sup_D \frac{i(C,D)}{\ell(D)}$$

alternate forms of subconvexity statement

Conj p, q Hölder conjugate

then

$$E_p(C) = \sup_D \frac{i(C,D)}{E_q(D)}$$

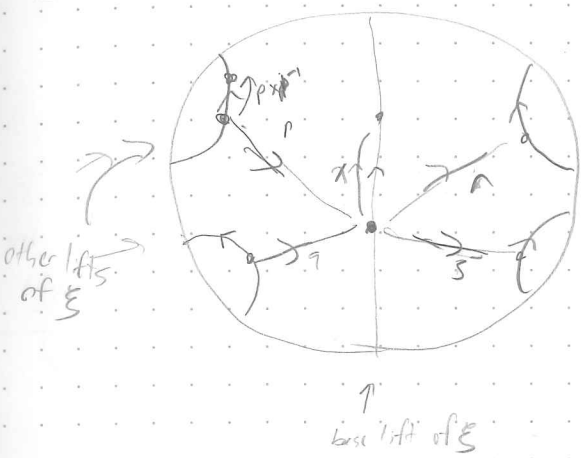
special case - extremal length

How is that proved?

Claim 1 $E_p(C)\, E_q(D) \gtrsim i(C,D)$

That's probably true. But not sharp unless C is simple (eg EL case)

Group elements for tori



Note "Base lift" could be far out of position.

This is not what we had before.

Oh — in group theory terms, probably $a = sp$, etc

Other elements:

$a = ps$
$b = qr$
$c = pr$
$d = qs$

} translations:

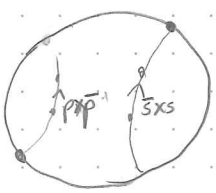
$a_n = px^n s$
$b_n = qx^n r$
$c_n = px^n r$
$d_n = qx^n s$

→ don't use this

Siddness constraints

$[ps] \cup [x] \searrow [pxs]$ "$[ps]$ crosses $[x]$" (and so on)

Should be equivalent to:



axes of pxp and sxs do not cross

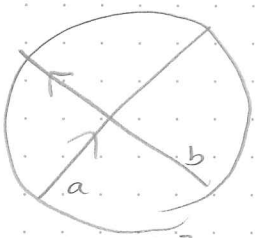
Why? is this equivalent?

(Maybe stronger than that?)

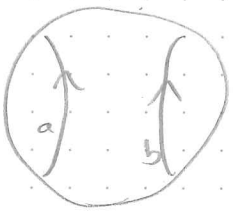
Generalities on smoothings

One of $[a] \cup [b]$, $[ab]$, $[aB]$ has a crossing and smooths to the others.

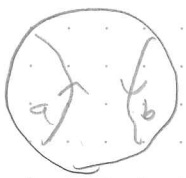
To see which, look at axes of $[a]$ and $[b]$ in $\mathbb{R}^2$:

(I) 

$[a] \cup [b]$ big

(II) 

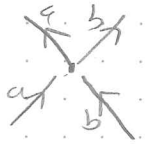
$[ab]$ big

(III) 

$[aB]$ big

or by considering order of geodesic reps of elts of π_1 [using canonical cyclic order on π_1]

(I) 

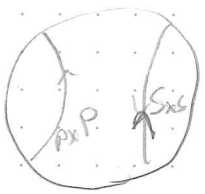
(II) 

(III) 

Note conjugation inv!

In example if $[ps] \cup [x^2] \rightsquigarrow [pxxs]$ ... ??

Hmm, Not getting desired effect.



$\rightsquigarrow$ [pxPSxS] big

Hmm, seems like there's lots of equivalent conditions

Given $a, b \in \pi_1(S)$

Cover corresponding to $\langle a, b \rangle$ is homeomorphic to

[careful]
both punctures easily, circles can be amphichiral

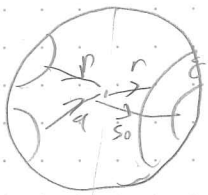

or


Claim.

$[a x A' x] \searrow [x] \cup [A x a] = 2[x]$

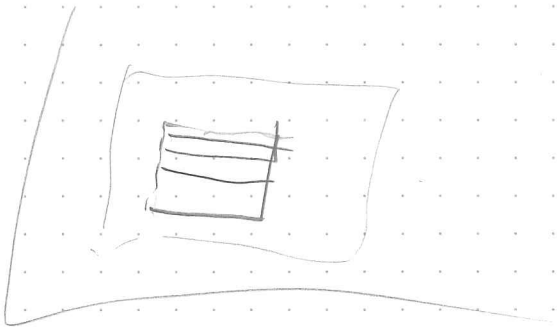
$\overset{?}{\Leftrightarrow} [a][x] \searrow [ax]$??

In any case. Need to worry about the "narrow window" problem



$s_1 = s_0 R x r$

$s_\wedge = s_0 R x^\wedge r$



$$\lim_{k\to\infty} \lim_{n,m\to\infty} v(rqx^n) + v(s_0 R x^k r p x^m) \to v(rpx^n) - v(s_0 R x^k r q x^m)$$

Eq. want this to be 0.

Should work.

After mix-and-match:

$$\lim_{n\to\infty} (rqx^n - x^n) + (s_0 R x^k - x^k) + (rpx^m - x^m) - (rpx^n - x^n) - (s_0 R x^k - x^k) - (rqx^m - x^m)$$

Order of limits doesn't matter.

Note the actions of $\pi_1(\Sigma, *)$ on lifts of $*$ to $\tilde{\Sigma} = \{\text{paths from } * \text{ up to } \sim\}$

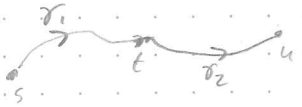
(a) precomposition with γ

This gives an isometry of $\mathbb{H}^2$

(b) postcomposition with γ

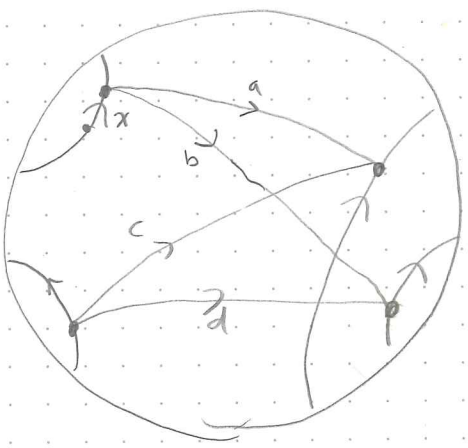
Not an isometry. Points move a bounded amount.
Only acts on lifts of $*$, not all of $\tilde{\Sigma}$

Convention: Compose paths in obvious way:

$\gamma_1 \cdot \gamma_2$ is 

so (a) is a left action, (b) is a right action

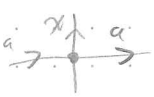
Setting of windows:



These are (b) pictures, post-composition w/ basepoint on fixed geodesic x

Constraint

- $aCdB = 1$
- a, b, c, d all cross x from R to L

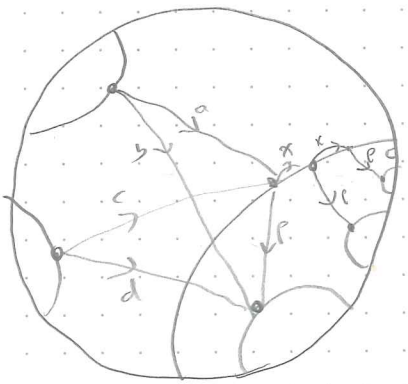


$[bc]$ smooths to $[a] \cup [d]$

To get the descending nested rectangles:

Translate one of the rectangles:

[after choosing a basic rectangle with this configuration, with 3 out of 4 endpts fixed]



set $e = Ab = Cd$

sequences: $\{(a, ae, c, ce)$ $(a, axe, c, cxe) \cdots (a, \underbrace{ax^k e}_{b_k}, c, \underbrace{cx^k e}_{d_k})$
($ae = b$, $ce = d$)

so interested in

$$\lim_{k\to\infty} \lim_{n\to\infty} v(\underbrace{ax^k e}_{b_k} x^n c x^m) - v(ax^n) - v(cx^k e x^m)$$

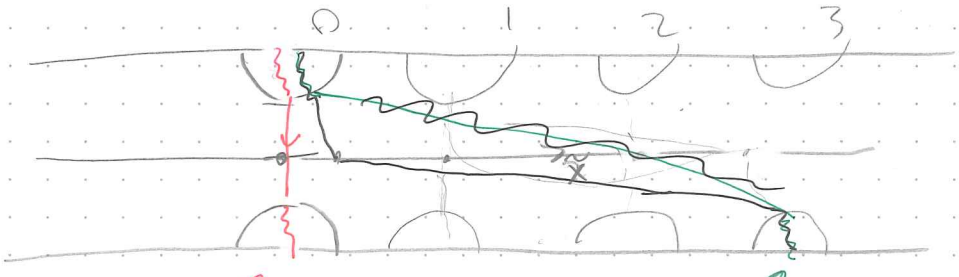
seems this should be 0 by the same techniques

Date: 2019 · 03 · 01

Smoothings (cont from 2018-10-03 #1)

Given: a, b crossing x to the right

Put in band model:



lift of a
(after normalizing numbering)

lift of b
illustrated: k = 3

k: chosen so that
a and $b x^k$
go to same lifts

$[a x^0][b x^0] \searrow \begin{cases} [a x^2 \; b x^{-2}] & r=2 \\ [a x \; b x^{-1}] & r=1 \end{cases}$

$[a x^n][b x^m] \searrow [a x^{n+r} \; b x^{m-r}]$

if $1 < r < (m-k) - n$

or $-1 > r > m-k-n$

$|r| > 1$

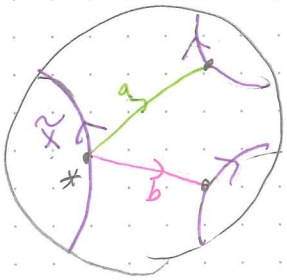
$|m-k-n|$

Lifts of ax^n, bx^m

x: simple

a, b: right across x.

(I)



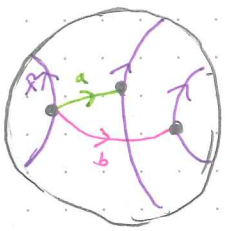
Three ~~two~~ cases:

(I) disjoint

(II) nested

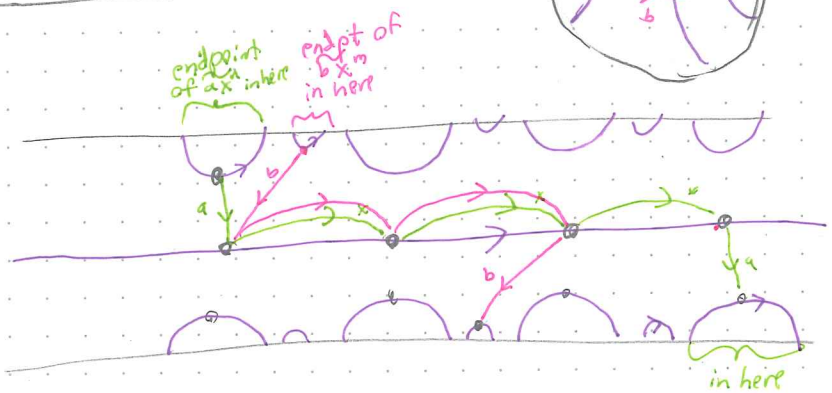
(III) equal: $b = ax^k$
↳ treat like nested

(II)



In band model

(I)



Oh, more than two cases:

- may be nested/disjoint on each side
- need to consider relation to other lifts of x

But in any case, each endpoint of $\tilde{a}x^n$ is constrained to lie in an interval of length < 1, in band model where translation by x is shifting 1 to the right.

Dynamics on geodesic currents

Can I define geodesic currents by intersection # with closed curves?

- We proved map was injective.

What about topology?

Eg Can we see action on filling currents is properly discontinuous?

Simpler Intersection # of two currents?

Easy bound filling currents wrt each other (?)

Q what's "filling"?

- Lower bound on $f(c)$ c an ss essential curve

Q What's a lamination? A $f(X) = \max(f(\gamma), f(c))$

Real question. How to define extension of τ_{cms} to "geodesic currents" in this sense?
- What's the topology on space of "geodesic currents"?

Idea instead of using this as a new def of geodesic current, use both notions. Some things will be easier in one notion or other, at least with proper topology.

Eg proper discontinuity of action on Riling currents looks easy in subspace of product topology.
Is that same as topology on geodesic currents?
- Also interested in action of MCG on space of measures on currents. This seems harder to see — what's a measure on a current in this language?
↳ But it's easy to give an exhaustion (compact) of space of currents

Page: 1

Date: 2019·04·15

Idea: Define space of "nearly convex" functions satisfying (quasi-convex??)

(a) sub-additivity $f(c_1 \cup c_2) \leq f(c_1) + f(c_2)$

(b) quasi-smoothing ~~$f(X) \geq f(\partial C) - K$~~

(c) stability

Consider dynamics of MCG on this space. $NC(S)$

Hope the only invariant, locally finite measures on nearly convex fcns are supported on measured laminations.

(and then there are classification results there, picking out Thurston measure on ML)

This should give counting results.

One strategy: start with initial nearly convex f

Consider $\lim_{L\to\infty} \frac{1}{L^{6g-6}} \sum_{\phi \in MCG} \delta\left(\frac{1}{L}\phi(f)\right)$ σ_L

Note this is apriori different from statement above. But feels very similar

show • limit exists [at a sublimit, at first]

• limit ~~has~~ is supported on fcns with ~~upgraded~~ properties:

- smoothing $f(X) \leq f(\partial C)$
- tree-like $f(X) = \max\left(f(\partial C), f(\widetilde{\ })\right)$
- additivity $f(c_1 \cup c_2) = f(c_1) + f(c_2)$

In all cases show properties of any limit by using a compact exhaustion of $NC(S)$

[Hmm, by what? Maybe fix filling curve γ. Look at $B_T(\gamma, L) = \{f \mid f(\gamma) \leq L\}$. Is that compact?]

should work for $\log$ enough

For each $f \in B_T^{n \text{ sat}}(\tau)$ (which is compact)

show that, eg, quasi-smoothing constant is small

and so anything in support of the limit must satisfy smoothing

- for "free-lice", want a lemma:

Prop (???) for any nearly-convex f, $\exists K_2$

so ~~$\max(f(x), f(\tilde{x})) \leq f(X) =$~~

(**) $f(X) \leq \max(f(x), f(\tilde{x})) + K_2$

adding a kind of symmetry. Want that to work with geodesic currents on hyp. groups in general anyway; combined they say

(**) Maximum two values of $f(X)$, $f(\tilde{x})$, $f(x)$ are within K of each other.

- Is there a natural conjecture to take care of ~~additivity~~? Is result even true? Recall closure of EL fcns in space of fcns on simple closed curves is really complicated

Q. ~~Is con~~ In $i(\cdot, \mu)$ for a geodesic current μ, is const in (**) related to $i(\mu, \mu)$?

Both are 0 at this same time. Scaling is different. -but that's not going to help in non-additive case.

Computing n_γ

To find n_γ, need to find region: $\{\mu \in \mathcal{ML} \mid i(\mu, \gamma) \leq 2\}$ and find the volume

Can write as a "max polynomial":

$$i(\mu, \gamma) = \max_i i(\mu, \gamma_i)$$

where γ_i are simple weighted multi-curves

That feels like it ought to help.

Could find the volume by finding $i(\mu, \gamma_i)$ explicitly in (say) DT coords for μ

Is there a more clever point of view?

Page: 1

Date: 2019·04·18

To show f positive on every quasi orthogonal surface

(sub-) additive, (quasi) convex

Then f is quasi s-convex:

$$f(X) \leq \max\left(f(\cdot), f(\widetilde{\cdot})\right) + k$$

NB

Requires symmetrical to make sense

why should this be?

False on torus (for subadditive)

and thus if filling/positive condition is dropped

How is positivity going to come in?

The "max" seems rather awkward.

Smoothings

(1) Limit $f_n = \frac{f(R^n)}{n}$ exists
(2) $f_n = f$ on 5-functions
(3) f_n continuous

---

(1) sub additivity

smooth $m^n(x) \sqcup m^m(p^n(x)) \longleftrightarrow m^{n+m}(x)$

Implies sub-additivity, then have Fekete's Lemma

[statement: ~~Subadditivity~~ Lemma extends

(2) Extension

→ argument with covering segments, complicated applications of Join lemma

Join lemma

(3) continuity

Superadditivity $f^n \gtrsim f^n$ in addition to other way

What's the natural order?

(1) Join Lemma statement
[May be omitting precise estimates.]
(2) smear it to get subadditivity
(3) Limit exists
(4) Limit continuous
(5) Limit extends

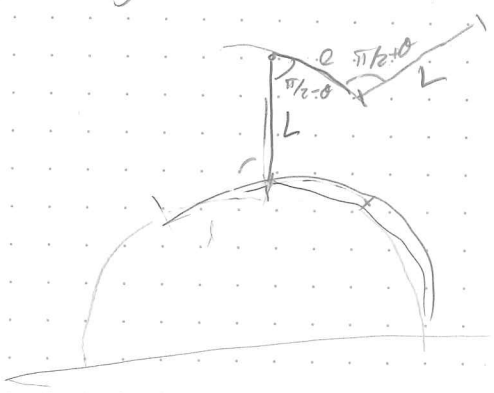
Then
Join Lemma proof can create curve between Lemma.

Page: 1

Date: 2019.05.10

Hyperbolic geometry

Landing point of broken path



transform for "furthest right" endpt:

$$\left(\quad \right) \left(\begin{matrix} \cos\left(\frac{\pi}{4}+\frac{\theta}{2}\right) & \\ - & \end{matrix} \right) \left(\begin{matrix} e^{L/2} & 0 \\ 0 & e^{-L/2} \end{matrix} \right)$$

$$Rot\left(-\tfrac{\pi}{2}+\theta\right) Trans(e) Rot\left(\tfrac{\pi}{2}+\theta\right) Trans(L)$$

⇝ get some matrix

Interested in the eigenvectors?

seems like a mess

Hyperbolic rotation matrix:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \quad \\ \quad \end{pmatrix}$$

$$\frac{ai+b}{ci+d} = \frac{(b+ai)(d-ci)}{c^2+d^2}$$

$$= \frac{bd+ac+(ad-bc)}{c^2+d^2}$$

$$\begin{pmatrix} \cos\theta/2 & \sin\theta/2 \\ -\sin\theta/2 & \cos\theta/2 \end{pmatrix}$$

Broken path lemma

Given: • "long" segments, length $\geq L$
• "short" segments, moving to the right by length in (ℓ_1, ℓ_2) (allowing negative amounts, meaning moving to the left)
• Angle between them in $(\frac{\pi}{2} - \theta, \frac{\pi}{2} + \theta)$

$\Gamma(L, \ell_1, \ell_2, \theta) \rightsquigarrow$ space of paths with these params

[Hmm, too many parameters]

Lemma A ~~For fixed ℓ_1, ℓ_2, θ and~~

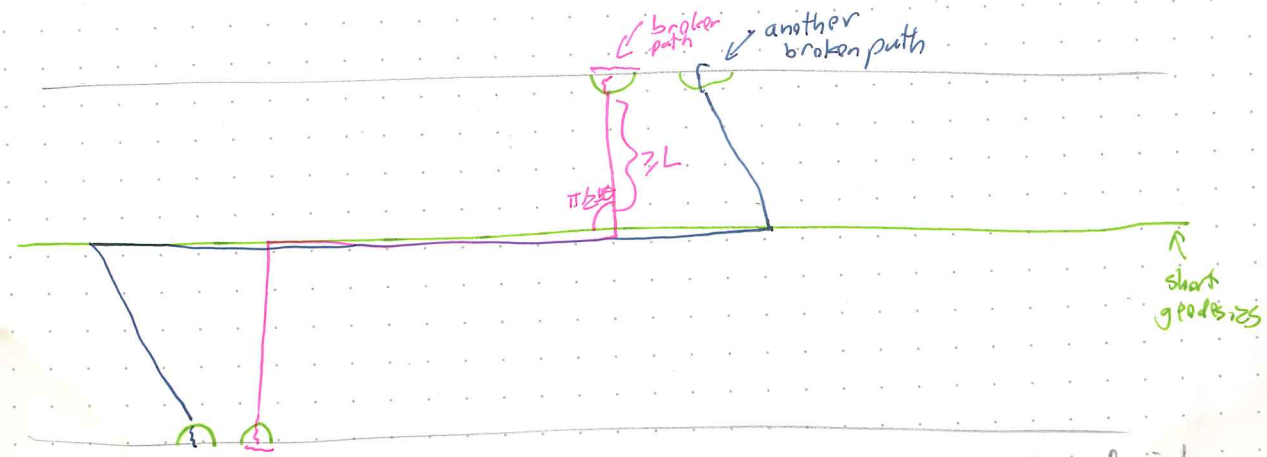
For broken paths γ with an upper bound on length of short segments and angle θ, ~~γ intersects~~ $\exists L$ so that for $\gamma \in \Gamma(L, \ell_1, \ell_2, \theta)$, γ intersects essentially geodesic containing any of the short segments.

Lemma B (Broken paths intersect).

What's the statement? It's not enough to talk about containment of intervals not without an assumption on θ

Broken path in bands model

(assuming L is large enough ~~for~~ for Lemma A)



For fixed θ, to guarantee no intersection, need the "long" endpoints to be distinct

Need Semi-ideal triangle

Hyperbolic law of cosines

$\cosh a = \cosh b \cosh c - \sinh b \sinh c \cos\alpha$

$\cos\alpha = -\cos\beta\cos\gamma + \sin\beta\sin\gamma\cosh a$

If Partially ideal: $\alpha = 0$

$$\frac{1+\cos\beta\cos\gamma}{\sin\beta\sin\gamma} = \cosh a$$

should get $a = 0$ when $\beta+\gamma=\pi$, yup

$$s = \frac{\cos(\beta+\gamma)+1}{\sin\beta\sin\gamma} + 1$$

$\cos(\beta+\gamma) = \cos\beta\cos\gamma - \sin\beta\sin\gamma$

case of interest: $\beta = \gamma = \frac{\pi}{2} - \theta$

$$\cosh a = \frac{1+\sin^2\theta}{\cos^2\theta} = \frac{2}{\cos^2\theta} - 1$$

$\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1$

$$= \frac{1+\frac{1}{2}(1-\cos 2\theta)}{\frac{1}{2}(\cos 2\theta + 1)}$$

Angle of parallelism

$\beta = F(a) = \text{or } \Pi(a)$

satisfies

$\sin\beta\cosh a = 1$

Right, this will work.

What about issue of getting a global cross-section? (that is an embedded disk)

Seems like a key question.

Hmm I think we're trying to be too precise

Find any global cross-section, approximately

Smoothings

$$f^k(\tau, \mu) = \int_\tau f([M^k(x)])\,\mu(x)$$

$$f^{k+\ell}(\tau, \mu) = \int_\tau f[M^{k+\ell}(x)]\,\mu(x) \le \int_{\tilde\tau} f[M^k(x)]\,\mu(x) + \int_{\tilde\tau} f[M^\ell(x)]\,\mu(x)$$

$$+ \int_\tau f(K)\,\mu(x) \to f(K)\,\mu(\tau)$$

Lemma ~~$R(\tau) = \mu$~~

$P_*\mu = \mu$ [needs defining] [related to extension argument]

Right: Lemma 8.1 is a special case of this.

To prove

$\Psi\mu$ is invariant under P

That is: $P: \mathcal{T} \to \mathcal{M}(\mathcal{T})$

upgrade $\mathcal{M}(\mathcal{T}) \to \mathcal{M}(\mathcal{M}(\mathcal{T})) \to \mathcal{M}(\mathcal{T})$

$$P(x) = \Psi(\varphi(x)) + (1 - \Psi(\varphi(x)) \cdot P(\varphi(x))$$

Know μ invariant under φ

$$P_* \mu = \Psi(\varphi(x))\, \varphi_*(\mu) + (1 - \Psi(\varphi(x))\, P_*(\varphi_* \mu)$$

($\varphi_*(\mu) = \mu$, $P_*(\varphi_* \mu) = \mu$)

~~$P_*(\Psi \cdot \mu) = \Psi(\varphi(x))\Psi(x)\,\mu$~~

$$= \Psi(\varphi(x))\,\mu + (1 - \Psi(\varphi(x)))\, P_* \mu$$

$\Psi(\varphi(x))\, P_* \mu = \Psi(\varphi(x))\, \mu$ → That's not right

$P_* \mu \in \mu$ since $P_*(\Psi\mu) = \Psi\mu$

$P_* \mu = \Psi\, P_* \mu$

R2Y7 In general $T: X \to Y$ μ measure on X

$(T_* f\, \mu)(S)$ T one-to-one

$$= \int_{f^{-1}(S)} f(x)\, \mu(x)$$

$$= f(T^{-1}(x))\, T_* \mu$$

$T^* f\, \mu = \boxed{f(T^{-1}(x))\, \mu(x)}$

$P^*\mu = ?$

$P_*\mu(S) = \int$

$P(x) = \psi(\varphi(x)) \cdot \varphi(x) + (1-\psi(\varphi(x)))\, P(\varphi(x))$

$G = \psi(F(x)) \cdot T(x)$

$G_*\mu = \int_S f(x)\, \mu(T^{-1}(x))$

$= f(x)\, G_*\mu$

$P_*\mu(x) = \psi(x)\, \varphi_*\mu(x) + (1-\psi(x))(\varphi_* P)_*(\mu)$

Want $P_*(\psi\mu) = \psi\mu$

$P_*(\psi(x)\mu(x)) = \psi(\varphi^{-1}(x))\, P_*\mu(x) = \psi(x)\mu(x)$

s.o want $P_*\mu(x) = \dfrac{\psi(x)}{\psi(\varphi^{-1}x)}\,\mu(x)$ wait, that's not finite

Ah, so confusing.

---

flux μ_T : transverse measure on cross-section

invt under φ (non-smeared return)

wts: $\psi\mu_T$ is invt under smeared return

$\psi = \int_0^1 \chi(\varphi^{-t}[0,t])\, dt$

Terminology & look at a paper

$P_\psi = \int P_{X_t}\, dt$

Hominid Return time for recurrent points is bounded below

↳ forward progress?

Possibilities for ~~constructing~~ global cross-section

These definitions for P_ψ aren't actually equivalent:

$$R_1(x) = \begin{cases} x & x \in \tau_0 \\ \psi(x)\, x + (1-\psi(x))\, R(p(x)) & x \in \tau_1 \setminus \tau_0 \end{cases}$$

$$P_1(x) = R(p(x))$$

$$P_2(x) = \int_0^1 \rho_t(x)\, dt = \sum \pi(a_i - b_i)\, y_i \qquad (*)$$

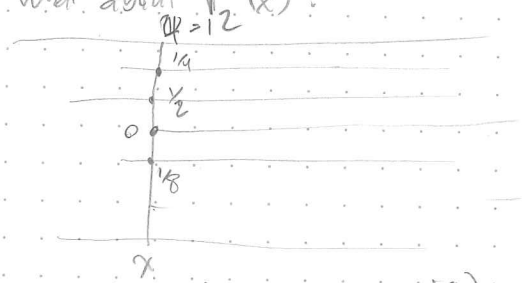
y_i = orbit of π
$a_i = \psi(y_i)$
$b_i = \max_{j<i} a_j$

where ρ_t is return to $\tau_t = \psi^{-1}([1-t, 1])$

---

are they both continuous? original $P(x)$ is. (Lemma 5.5)

What about $P_2(x)$?



Looks continuous by $(*)$ (since a_i, b_i are).
But not smooth.

so P_1 seems nicer from that point of view, though we don't use smooth in this paper

Less If my flux for τ_1 is ψ_M a flux for P_1? Integral proof does not apply

Date: 2019 . 05 . 23

To see $\psi(x)$ is invariant under $P(x)$

$$R(x) = \begin{cases} x & x \in \tau_0 \\ \psi(x)\, x + (1-\psi(x))\, R(\rho(x)) & x \in \tau_1 \setminus \tau_0 \end{cases}$$

"forward orbit"

$$P(x) = R(\rho(x))$$

$$R(x) = \psi(x)x + (1-\psi(x))\,\psi(\rho(x))\,\rho(x) + \cdots$$

$$= \sum_{k=0}^{N} \rho^k(x) \cdot \psi(\rho^k(x)) \,(1-\psi(\rho^{k-1}(x))) \cdots (1-\psi(x))$$

$$P(x) = \psi(\rho(x))\rho(x) + (1-\psi(\rho(x)))\,\psi(\rho^2(x))\,\rho^2(x) + \cdots$$

$$(\lambda x.\ \psi(x)x)_*(\mu) = \psi\mu$$

~~$(\lambda x.\ f(x)T(x))_*(\mu)(S) := \int_{x\in S}$~~

$$T_*(f\mu) = f(T^{-1}(x))\,T_*\mu$$

$T: X \to Y$

$$T_*\mu(S) = \mu(T^{-1}(S))$$

$f: X \to \mathbb{R}_{\geq 0}$

$$(fT)_*\mu(S) = \int_{T^{-1}(S)} f(x)\,\mu(x) = \int_X f(x)\,\chi_S(T(x))\,\mu(x)$$

← That's not a bad expression

T invertible

$$(fT)_*\mu = \int_X = \int_Y f(T^{-1}y)\,\chi_S(y)\,(T_*\mu)(y) = \boxed{f(T^{-1}(y))\,T_*\mu}$$

$$R_*\nu = \psi(x)\nu(x) + (1-\psi(\rho^{-1}(x)))\,\psi(x)\,\rho_*\nu(x) + (1-\psi(\rho^{-2}(x)))\,(1-\psi(\rho^{-1}(x)))\,\psi(x)\,\rho_*^2\nu(x) + \cdots$$

$$P_*\nu = \psi(x)\rho_*\nu(x) + (1-\psi(\rho^{-1}(x)))\,\psi(x)\,\rho_*^2\nu(x) + \cdots$$

$$P_*(\psi\mu) = \psi(\rho^{-1}(x))\,\psi(x)\,\mu(x) + \psi(\rho^{-2}(x))\,(1-\psi(\rho^{-1}(x)))\,\psi(x)\,\mu + \cdots$$

Telescopes as desired

since some $\psi(\rho^{-k}(x)) = 1$

smoothings : Continuity

~~Want~~

Have $f_\tau(\mu) \le \frac{f_\tau^k(\mu) + k(\tau)\mu(\tau)}{k}$

Want some lower bound on $f_\tau(\mu)$

Dilac. says $2f_\tau^k(\mu) - f_\tau^{2k}(\mu) \le k(\tau)\mu(\tau)$ [Probably need some constants here]

Proof $M^{2k}(x) \cup K \searrow M^k(x) \cup M^k(P^k(x))$ [Join Lemma (b)]

$f(M^k(x) \cup M^k(P^k(x))) \le f(M^{2k}(x)) + k(\tau)$

integrate both sides with ψ_μ

$f_\tau^{2k}(x) + f(k)\mu(\tau) \ge \int_x f(M^k(x))$

Use fact $\int_x M^k(x)\psi_\mu = \int_x M^k(P^k(x))\psi_\mu \overset{\text{def}}{=} M^k(\psi_\mu)$

So $M^{2k}(\psi_\mu) \cup K \searrow 2M^k(\psi_\mu)$

$f_\tau^{2k}(\mu) + k(\tau)\mu(\tau) \ge 2f_\tau^k(\mu)$

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Date: 2019.06.04

From curves to currents

I. Why currents? Def, examples

II. Statement, comparison, analogues

III. Smeared return map

Joint w/ Didac Martinez Granado

Spring 2022 MSRI program

Complex Dynamics: From Special Families to Natural Generalizations

Koch, Raissy, Schleicher, Shishikura, Thurston (fully/geo)

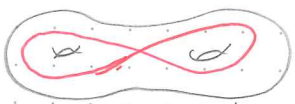
Companion program (IP) Analysis & Geometry of Random Spaces

Bonk, Viklund, Lind, Rohde, Saksman

Topological objects on surfaces



homotopy classes of simple closed curves



general multi-curves

[or maybe 4-punctured sphere?]



measured [singular] foliations



geodesic currents

$S(S) \hookrightarrow C$

$\downarrow \int \qquad \downarrow \int$

$ML \hookrightarrow GC$

Introduced by Bonahon.

Lots of geometry, indep of hyp. structure

- Every hyp surface Σ has a Liouville current L_Σ
- geometric intersection # extends continuously to currents
- $i(L_\Sigma, c)$ = hyperbolic length of c

simpler than in Prof. Sibony's talk: 1D geodesic current, so can lift to T^1

Formally: Measures on $T^1\Sigma$ invariant under geodesic flow, weak* topology

Counting results

Rafi-Souto 2012 (building on Mirzakhani)

Thm $f: GC(S) \to \mathbb{R}_{>0}$
α a filling current

$$\lim_{L\to\infty} \frac{\#\{\varphi \in \mathrm{Map}(S) \mid f(\varphi(\alpha)) \le L\}}{L^{6g-6}} = \frac{m(\alpha)\, m(f)}{m_g}$$

(collision of motivation)

Ex # curves $\le$ given length of given (filling) top. type

Applies for any function that extends to currents. which includes

- intersection #s
- hyp. length
- length wrt non-pos metric
- length wrt simple/general generating set

} Each a paper

~

Thm (Martinez-Granado, T)
$f: C(S) \to \mathbb{R}$ a function (weighted multi-curves) that satisfies

(1) (quasi) smoothing: $f(X) \ge f(C)$ $(+C)$

(2) concave union: $f(C_1 \cup C_2) \le f(C_1) + f(C_2)$

[(3) homogeneity: $f(kC) = kf(C)$]

[(4) stability: $f(C^k) = kf(C)$]

then V_f (stabilization of f) extends continuously to currents.

(& thus have counting)

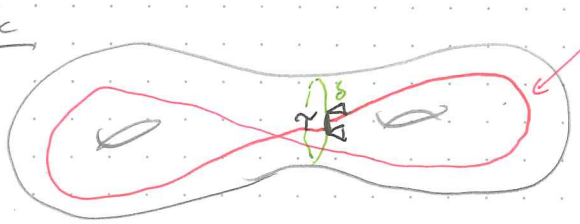
New example $\sqrt{\text{Extremal length}}$ ← Example Length wrt any Riemannian metric

Think Concavity $\Rightarrow$ continuity

Constructing extension Recall geodesic flow φ_t on $T^1\Sigma$ to φ_t

Pick a global cross-section τ that is simply-connected, nearly $\perp$ to a geodesic δ

Think



[but this is inaccurate; need to intersect every closed geodesic]

geodesic current α induces flux $\alpha_\tau = \mu$ on τ

[Note $T^1\Sigma$ a 3-manifold, τ a 2-manifold w/ boundary]

construct curves $m^1(x), m^2(x), \dots$ on Σ by following first return map and closing up along τ

$p : \tau \to \tau$
$m : \tau \to \pi_1(\Sigma)$

Def $$f(\alpha) = \lim_{n\to\infty} \frac{1}{n} \sum_{\vec{x}\in\tau} f(m^n(\vec{x}))\, \alpha_\tau(\vec{x})$$

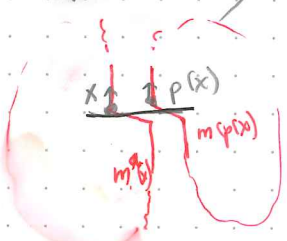
Example Suppose α is geodesic current from a closed curve C that intersects τ in $\vec{x}_1, \dots, \vec{x}_k$

Then $m^{ak}(\vec{x}_i) = C$

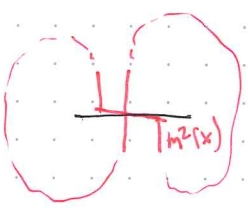
$$\frac{1}{ak} \sum_i f(m^{ak}(x_i)) = \frac{1}{ak} \sum_{i=1}^{k} f(C^a) = f(C)$$

Why does the limit converge?

~~Need to say something about behaviour~~



vs



Looks like you can smooth from one to the other.
But how to make sure? Add copies of δ use hyp. geom.

- guarantee "geodesic" part is long relative to length of δ
- smooth with several copies of δ
- make connected
- smooth with several copies of δ oriented other way



$\rightsquigarrow$ ~~for~~ for sufficiently large a, b

$$K_i^2 + \left| f(m^a(x)) + f(m^b(p^a(x))) \geq f(m^{a+b}(x)) \right| + C$$

Fekete's lemma: Limit converges

What about continuity? Weak* topology - want to integrate μ against continuous functions. Return map is not continuous

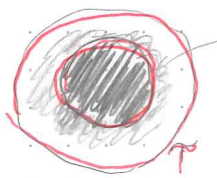
A Continuous (smeared) return map

$\tau_0 \Subset \tau$ compactly contained pair of global cross-sections

$\psi : \tau \to [0,1]$ bump function 1 on τ_0, 0 on $\partial\tau$

Smeared return map $P: \mathcal{T} \to \mathbb{R}_1\mathcal{T}$ ← linear combs w/ total weight 1

$$P(x) = \begin{cases} p(x) & x \in \mathcal{T}_0 \\ \psi(p(x)) \cdot p(x) + \bar{\psi}(p(x)) P(p(x)) & x \notin \mathcal{T}_0 \end{cases} \qquad \bar{\psi} = 1 - \psi$$



$M \circlearrowleft \mathcal{T} \to \mathbb{R}, \pi_1(\Sigma)$ similar

we actually use this smeared return!

$$\hat{f}_{\psi\mathcal{T}}(\gamma) = \int_{x \in \mathcal{T}} f(M^n(x)) \psi(x) \mu(x)$$

linear combination of curves, depending continuously on x

$$f(\gamma) = \lim_{n \to \infty} \frac{f^n_{\psi\mathcal{T}}(\gamma)}{n}$$

Get uniform convergence:

$$\left| f(\gamma) - \frac{f^n(\gamma)}{n} \right| \leq \frac{K(\psi\mathcal{T}) \mu(\psi\mathcal{T})}{n}$$

Further bits:

- Hyp. geom. estimates
- stability: $\|f\|(c) = \lim_{k \to \infty} \frac{f(kc)}{k}$ (dropping ~~homogeneity~~ hypothesis) f defined only on curves

~~to get~~ Eg length wrt arbitrary gauge

to get quasi-smoothing: $\exists C \searrow_1 C' \Rightarrow kC \searrow_k kC'$

Suggestions

- self-intersection #?
- apply to other flows, like Lorenz attractor?
- theory for surfaces (Thurston norm)??
- Punctured surfaces?

Q What is natural metric on geodesic currents?

Wasserstein distance?? → Need a metric on surface for that

Don't necessarily expect a natural one or any invariant measure on GC, other than this! supported on ML

TODO: U options → external tensor prod of type D structs

Roeder "split diamond" : recall the graph

- never use pure green (or other colors), always mix w/ black

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11 (2016-06-06

Erasing points

Given $\varphi: \Sigma_1 \hookrightarrow \Sigma_2$ [maybe don't need inclusion?]

geodesic current ν on Σ_1

Want to define $\varphi_* \nu$

Let's ignore subtleties with which curves, say by assuming Σ_2 is closed

For c a curve on Σ_2, want to define $i(c, \varphi_* \nu)$ as an infimum of some sort

Need seems plausible. Then you have

- additivity
- smoothing

⇒ have current $\varphi_* \nu$

Is this "lower semi-continuous"? What does that mean? Something about values of curves?

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Q. Can geodesic currents technology be used for easier proof of ending laminations conjecture? [Refs Bonahon '86]